Same-Sex and Opposite-Sex Couples and the Child Penalty

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Abstract

Existing work has shown that the entry of a child into a household results in a large and sustained increase in the earnings gap between male and female partners in opposite-sex couples. We expand this analysis of the child penalty to examine within-couple dynamics in earnings for both opposite-sex and same-sex couples in the U.S. around the time their first child enters the household. Using linked survey and administrative data and event-study methodology, we confirm earlier work finding a child penalty for women in opposite-sex couples. We find this is true even when the female partner is the primary earner pre-parenthood, lending support to the importance of gender norms in opposite-sex couples. By contrast, in both female and male same-sex couples, earnings changes associated with child entry differ by the relative pre-parenthood earnings of the partners and tend towards equalization: secondary earners see an increase in earnings, while primary earners see a small decrease.

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1 Introduction

When a woman living in the United States has a child, her earnings fall by on average 21-61% depending on her state of residence (Kleven, 2022). This phenomenon, often referred to as the "child penalty," has been observed in various countries and is significant and persistent even in places where safety nets and job protections are robust (Kleven et al., 2021). Although the child penalty results in a large and sustained increase in the earnings gap within opposite-sex couples (Angelov et al., 2016; Chung et al., 2017), less is known about the child penalty for same-sex couples.[1] By comparing the earnings dynamics of opposite-sex and same-sex couples around the time their first child enters the household, this paper aims to characterize the earnings dynamics of same-sex parents as well as to understand the extent to which the child penalty is driven by earnings or work-life preferences versus by societal gender norms or equality norms that impact the labor and home production decisions when childcare responsibilities are introduced.

Our main analysis characterizes how individual earnings change after same-sex and opposite-sex couples become parents for the first time. We start with the simplifying assumption that work-life preferences of same-sex couples are comparable to those of opposite-sex couples so that we can concentrate on the gender norms of same-sex couples and those of opposite-sex couples. Our focus on within-household earnings dynamics allows us to explore how pre-parenthood earnings, as a measure of expected post-child earnings, are related to the labor/home production tradeoffs faced by the partners. We investigate whether one member of a same-sex couple appears to assume a primary caregiver role, resulting in an increase in the earnings gap after the child enters the household, or whether the additional demand on their time is more equally shared than in opposite-sex couples.

In addition to examining how individuals' earnings change, we also explore the effect of a

[1] Throughout the paper, we adopt the naming convention from the American Community Survey (ACS) of "same-sex" and "opposite-sex" couples. Thus our convention defines the experience, for example, of a gay male with a female partner as being part of an opposite-sex couple. We acknowledge that this convention cannot capture the full range of complex identities and behaviors.

child on the combined earnings of the couple. Our goal is to determine whether household earnings trends change with the arrival of a child. Although most studies on the child penalty do not explicitly measure this, if male partners experience no change in their earnings after having a child and female partners experience a significant decrease, we would expect to see a decrease in household earnings in opposite-sex households. Our study explores whether this potential decrease in household earnings also exists for same-sex couples. Furthermore, the path of household earnings when a child enters has not been documented for same-sex couples in the U.S. and may depend on how the couple divides home and market production.

We use an event-study design to estimate the earnings trends around the entrance of the first child. Following the methodology of Kleven et al. (2019), Andresen and Nix (2022), and others, we compare earnings in quarters after the first child enters the household to earnings six quarters before the child enters. The timing of the event provides exogenous variation in parental status due to the exact quarter in which the child enters. Our identifying assumption is that the only source of any discontinuous change in earnings upon the child's arrival is the increase in required household production associated with the child. In our main specification, we use inverse propensity weights to ensure that same-sex couples and opposite-sex couples have similar background characteristics, enabling us to compare couples whose income trajectories would be the same if not for the introduction of a child.

Our main empirical analysis relies on a combination of Census Numident (Social Security) and Internal Revenue Service (IRS) 1040 tax returns to identify couples and their children. The combination of these two administrative datasets provides a large sample of same-sex couples, allowing us to carry out our analysis separately for female and male same-sex couples. In an alternative specification, we instead use data from the American Community Survey (ACS) to obtain measures of couples, their identifying characteristics, and their children. We supplement the ACS data with date of birth data from the Census Numident and IRS 1040 tax returns in order to be more precise about the date when an adopted child first enters the household. The ACS sample is smaller, but includes both spouses and unmarried couples, while our main IRS-Numident sample

only includes married couples. Despite those differences, we find that our results are consistent across the two samples. In both cases we define our outcomes of interest using individual quarterly earnings from the Longitudinal Employer-Household Dynamics (LEHD) dataset covering years 2005-2021. These data include quarterly earnings records collected by state unemployment insurance (UI) programs and provide coverage for over 95% of private-sector workers, as well as almost all state and local government employment (Abowd et al., 2009).

We find that the child penalty in opposite-sex couples applies almost exclusively to the female partner even if she was the primary earner before parenthood. In contrast, neither partner in same-sex male or same-sex female couples experiences as strong or persistent a drop in earnings as females in opposite-sex partnerships. If anything, the individual-level earnings dynamics of same-sex couples are suggestive of a trend of equalization of earnings post-child, with the primary earners experiencing no change in earnings and the secondary earners increasing earnings. However, we also see that among same-sex equal earners, there is a shift toward specialization after the child enters. We explore this further by comparing the gap in earnings within couples before and after becoming parents and find that across all couple types there are shifts toward specialization after the child enters. That specialization does not necessarily follow the patterns seen pre-child; after the child enters, some secondary earners shift to be primary earners, and vice versa in other couples. At the household level, we find that both opposite-sex and same-sex households experience a reduction in earnings in the year following the child's arrival, regardless of how the earnings losses are distributed across partners.

Disaggregating the results into intensive (earnings) and extensive (employment) margins of earnings changes, we see that the sustained losses incurred by women in opposite-sex couples primarily come from movement out of the labor force. For those who stay in the labor force, earnings recover within a year of the child's arrival. For men in opposite-sex couples, the upwards trend in earnings post-child is even greater among primary and equal earner men when only considering the intensive margin. In contrast, because secondary earner men in opposite-sex couples are more likely to be employed after the child enters than before, growth at the exten-

sive margin is larger for them than the growth at the intensive margin. For both male and female same-sex couples, we see growth in earnings at the intensive margin for primary and equal earners, while the trend in total earnings remains close to zero. For secondary earners in same-sex couples, the change in earnings at the intensive margin is very close to the overall change in earnings, suggesting that there is limited extensive margin adjustment among secondary earners in same-sex couples after a child enters the household.

To explore the dynamics we observe in the empirical analysis, we develop a simple theoretical model of the factors that influence the division of time between home production and paid work when a child enters a household. This model abstracts away from any differences in preferences for children and constraints across opposite-sex, male same-sex, and female same-sex couples. We explore three possible decision rules regarding the allocation of additional child-related home production work: allocation based on equality, allocation based on comparative advantage in earnings, and allocation based on societal gender norms. We contrast these scenarios with one based simply on maximizing total household earnings.

The paper is organized as follows: Section 2 provides a literature review and describes our contributions to that literature; Section 3 describes our data and methodology; Section 4 presents our results; Section 5 outlines our theoretical framework; and Section 6 provides discussion of the implications of our results and possible future research directions.

2 Literature Review

We contribute to two strands of literature by examining the earnings impact of the first child for same-sex (both female and male) versus opposite-sex couples in the United States. First, we provide a novel view of parental earnings penalties in the U.S., estimating effects for both members of the couple. Second, we contribute to the growing literature describing the characteristics of same-sex couples. Our paper builds on previous work combining these two threads that uses Scandinavian data; we analyze the earnings dynamics in the U.S. context, where the social safety net for new parents is more porous both with respect to parental leave and childcare. The

larger U.S. population also allows us to separately explore the patterns in earnings of same-sex male couples with children which was not possible in the Scandinavian countries studied.

Much of this work builds on insights from Becker's (1965) classic household comparative advantage model. According to this model, we would expect the partner with higher earnings to specialize in paid market work and the partner with lower earnings to assume greater responsibility for home production work. It becomes challenging to disentangle the effects of gender norms from those of comparative advantage when higher earnings tend to accrue to one sex.

Previous studies have shown that the earnings gap widens among individuals in opposite-sex couples when they have children (Angelov et al., 2016; Chung et al., 2017). Kleven et al. (2021) find that the child penalty exists even when couples adopt. Other studies focus on mothers and show that their earnings decline, presumably leading to an increase in the earnings gap within couples (Byker, 2016; Hotchkiss et al., 2017; Kleven et al., 2019; Kuziemko et al., 2018; Neumeier et al., 2018; Sandler et al., 2019). Kleven et al. (2019) show that opposite-sex couples in Scandinavian countries experience smaller child penalties than in the United States. Our analysis comparing patterns between opposite-sex couples and same-sex couples highlights the role of gender norms, which Kleven (2022) shows have an important role in the geographic distribution of the child penalty across the U.S. Using both spatial and temporal variation, he finds that gender norms, which may vary across localities, are an important determinant of the presence and magnitude of a child penalty.

Most of the evidence regarding the parental earnings gap for same-sex couples comes from Scandinavian countries. However, these studies have been unable to investigate male same-sex couples with children due to their small numbers. Therefore, this paper makes another important contribution by investigating parental earnings gaps among male same-sex couples. Andresen and Nix (2022) explore the differences between three groups of couples (opposite-sex adopting; opposite-sex not adopting; and female same-sex) to better understand the mechanisms underlying the child penalty. Using Norwegian data, they find that females in opposite-sex couples experience a substantial child penalty upon the birth of the child, but males in opposite-sex couples do

not. Both parents in female same-sex couples experience a child penalty, but it is larger for the birth mother. They conclude that the long-term child penalty is not driven by giving birth or male labor market advantage, since there is recovery in even the birth-mother's earnings in same-sex couples. Instead, they suggest that preferences, gender norms, and discrimination play a significant role.

A number of studies have identified characteristics of same-sex couples that lead us to anticipate differences in the child penalty between opposite-sex and same-sex couples, as well as to anticipate differences between male and female same-sex couples. Same-sex couples are more likely to assortatively match on some dimensions, leaving less opportunity for specialization within couple. For example, Ciscato et al. (2020) examine homogamy in couples over race, age, and education using 2008-2012 ACS data for California. They find that same-sex couples, especially female same-sex couples, are more likely to have similar educational attainment. Black et al. (2007) likewise note that, based on Becker's (1991) theory of specialization, "we expect to observe less household specialization in gay and lesbian couples" (p. 61). On the other hand, same-sex couples are less similar on other dimensions; Ciscato et al. (2020) document that they are more likely to be different races and ages than opposite-sex couples.

Relatedly, same-sex couples are more likely to have smaller earnings gaps compared to opposite-sex couples. Antecol and Steinberger (2013) and Jepsen and Jepsen (2015) use data from the 2000 Decennial Census to show that same-sex couples in the U.S. have smaller earnings gaps than opposite-sex couples, mostly due to differences in the presence and number of children. Black et al. (2007) find that same-sex couples are more likely to be dual-earner households than opposite-sex couples, though when they are not dual-earner households, the "stay-at-home" partner is more likely to be the one with lower education for both opposite-sex and same-sex partners. Both same-sex and opposite-sex couples are more likely to be single-earner households once they have children.[2] In terms of empirical evidence of specialization, Hofmarcher and Plug

[2]Additional evidence from Hansen et al. (2020) shows that marriage equality laws increased specialization within same-sex female households. Since married same-sex couples are more likely to have children than unmarried, that increase in specialization could be in line with Black et al.'s (2007) application of Becker's theory.

(2022) use the ATUS and find that opposite-sex couples tend to specialize more than same-sex couples. This difference is driven by strongly specializing couples where the female partner is out of the labor force. Giddings et al. (2014) use ACS and Decennial data and find that there is less specialization in same-sex couples as compared to opposite-sex couples, but this gap is shrinking over time.

Kurdek (1993) surveys about 300 *childless* couples in the U.S. and finds three patterns for the distribution of household tasks: same-sex female couples follow an "equality pattern" in which either partner is equally likely to undertake a task; same-sex male couples follow a "balance pattern" in which partners specialize in tasks that are equally distributed; and opposite-sex couples follow a "segregation pattern" in which one partner does most of the tasks (and that partner is the woman). Thus, same-sex couples distribute tasks more equally than opposite-sex couples but the equality is attained in male same-sex couples through specialization and in female same-sex couples by sharing or alternating tasks.

Traditional gender norms that place the time cost of childcare on the female partner will function differently, if at all, in same-sex couples. However, other norms may take their place, based on age, education, birth parent, genetically-related parent, or other ways of organizing family dynamics. For example, in female same-sex couples, there could be a norm that the birth mother takes on more of the childcare duties, as Andresen and Nix (2022) find in the short run for same-sex female couples in Norway.

Prickett et al. (2015) use the American Time Use Survey (ATUS) to examine how parents spend time with children. They find that women in both same-sex and opposite-sex couples and men in same-sex couples spend significantly more time with children than men in opposite-sex couples. Additionally, there may be a stronger culture of co-parenting in female same-sex couples. Genadek et al. (2020) use the ATUS to examine the amount of time couples spend together and find that female same-sex couples spend more time together overall and coparent more than opposite-sex couples, who in turn coparent more than male same-sex couples.[3] Using interna-

[3]Both studies using the ATUS caution about interpretation given the very small sample sizes for same-sex couples.

tional data and focusing on four household tasks (not including childcare), Bauer (2016) finds patterns similar to those in Kurdek (1993): female same-sex couples share housework tasks more than male same-sex couples and more than opposite-sex couples (the percent of shared housework tasks are 41%, 33%, and 28% respectively). However, Bauer finds the presence of children "traditionalize the domestic work pattern," that is, increase task segregation for both female same-sex couples and opposite-sex couples (but not for male same-sex couples).

Another difference between same-sex and opposite-sex couples is their family structure. Same-sex couples tend to have fewer children and the children they have are typically older. The children of same-sex couples are more likely to have come from earlier relationships (Genadek et al., 2020). Same-sex couples are also more likely to have adopted children or stepchildren than opposite-sex couples (Badgett et al., 2021). Walker and Taylor (2021) estimate that 17.2% of married same-sex couples (5.9% of unmarried couples) live in households with only adopted children, compared to 1.7% of married opposite-sex couples (0.6% of unmarried couples). Same-sex couples are also more likely to foster children, with 1.7% of married same-sex couple households (1.2% of unmarried partners) fostering children compared to 0.2% of married opposite-sex couple households (0.1% for unmarried partners).

3 Data and Methodology

We rely on four main data sources, the American Community Survey (ACS), Internal Revenue Service (IRS) 1040 forms, the Census Numident, and the Longitudinal Employer-Household Dynamics (LEHD) dataset. The ACS, IRS, and Numident files provide us with demographic and relationship information, while the LEHD provides us with administrative earnings used to construct our outcome variables. We link these various sources of data using unique identifiers called Personal Identification Keys (PIKs) to construct a comprehensive dataset, as we detail in Appendix A.1.

3.1 Primary Analysis Sample: IRS - Numident

We take two different approaches to identifying the couples used in our analysis, each with its own advantages and disadvantages. For our main analysis we rely on a combination of Census Numident and IRS 1040 files. Our approach is to identify individuals who are in couples as those who filed taxes jointly as determined from IRS data spanning 2013-2019.[4] We then use the 2020 Census Numident, which sources sex from Social Security Administration (SSA) administrative records, to define both the sex of the individuals and the date of birth of the child.[5] Thus "same-sex" couples are defined as those in which both individuals are male or both individuals are female and "opposite-sex" couples as those in which one individual is male and one is female. Identifying couples based on point-in-time tax filings means we cannot confirm that couples are together during the entire period we study. Our approach identifies individuals who were coupled at the time the child entered the household, but we cannot confirm whether their family arrangements changed in the future.

Basing our analysis sample on joint tax filers yields a large sample, particularly of same-sex couples, which make up a relatively small portion of the country's population. To avoid computational intractability, we use a 10% sample of opposite-sex couples to form our analysis sample. Table 1 provides the number of same-sex male, same-sex female, and opposite-sex couples in our IRS-Numident sample. We have 1.1 million opposite-sex couples, 99,000 female same-sex couples, and 46,000 male same-sex couples. This compares to about 10,000 same-sex couples (2,000 of which are male same-sex) that we are able to identify in survey data. As a result, same-sex couples make up a larger proportion in our anaylsis sample than they do in the overall population. In our sample, opposite-sex couples account for about 88.4% of couples, with 4% male same-sex couples and 8% female same-sex couples.

This analysis sample has the advantage of identifying a large group of same-sex couples,

[4] We require the household to have at least three people (partners and at least one child).

[5] Sex as used in this paper is measured by administrative records; any difference in identity based on self-reports is not captured in this paper. An excellent overview of current research into this measurement question is the National Academies of Sciences, Engineering, and Medicine report.

which allows us to tightly estimate the impact of first child on the earnings path. However, this approach limits the focus to married couples only, who we see on joint tax filings. In addition, we lack many other basic demographic variables, such as race and ethnicity, education, and occupation, which would allow us to better explore differences between the opposite-sex and same-sex couples in our analysis. Therefore we use a secondary approach to building our analysis file based on survey data. Our analysis using these two different ways of constructing the sample leads to very similar results.

3.2 Secondary Analysis Sample: American Community Survey (ACS)

As an alternative to our main adminstrative-data-based analysis sample, we use American Community Survey (ACS) data to identify couples. The benefit of this approach is that we can include both married and unmarried couples and we have more demographic information about the couples. However, our sample sizes are much smaller, especially for male same-sex couples (2,200 in the ACS sample versus 46,000 in the IRS-Numident sample). In addition, we are concerned that survey errors may lead to discrepancies between the survey-reported same-sex couple status and actual same-sex couple status. Kreider et al. (2017) note that "when two groups are related, and a very small proportion of the large group makes mismarks, their answers can affect the estimates of the smaller group" (pg. 2). For these reasons, we prefer our IRS-Numident sample and predominantly present results from analysis on that sample. However, we include a discussion of the ACS sample here and some results from this sample in the appendix.

The secondary sample comprises couples with children from the 2013-2019 ACS. The ACS is sent to approximately 290,000 addresses each month, totaling about 3.5 million addresses annually. Statistics from the ACS are currently published in annual formats of 1-year and 5-year estimates; we use the 1-year estimates for 2013-2019. The ACS collects detailed information for up to 5 people in the household and less detailed information for up to 7 additional household members. We identify couples and children in the ACS using question 2, which asks about the relationship to "Person 1." The question includes 15 checkbox responses for 2013-2018, with the

options "Husband or wife" and "Unmarried partner" for identifying couples. From 2019 onward, the relationship question responses are disaggregated into further categories: "Opposite-sex" and "Same-sex" (see Appendix A.1, and especially Figure AF1, for a detailed discussion of this question over time). Thus, prior to 2019 we determine "opposite-sex" and "same-sex" using the relationship question combined with the question about sex (question 3), while in 2019 we can simply use the relationship question.[6] In all years, we define a couple as "spouses" or "unmarried partners." This definition is based on a point-in-time indicator of couplehood, as we do not have a longitudinal measure of couple status. This means we cannot confirm that couples were together during the entire analysis period, or even at the time the child entered the household.[7]

The ACS relationship question includes four checkbox responses for children (biological, adopted, step, and foster children). In our analyses, we include biological, adopted, and stepchildren, but exclude foster children due to difficulties identifying when they enter and exit a household, as well as the different relationship dynamics involved. We use birth dates from the ACS to identify the timing of the child's birth for up to three children per household and validate those birth dates using the Census Numident.[8] To accurately identify when the child enters the household in the case of adoption, we use IRS 1040 data, using the year in which the child is first claimed by the head and/or partner as the year of entry for cases in which the child was adopted.[9]

3.3 Estimates of the Prevalence of Same-Sex Couples

According to data from the 2019 ACS, there are around 66 million couples in the U.S., with the majority being married couples (58 million as compared to 8 million unmarried partners) (Walker and Taylor, 2021). Of these 66 million couples, 65 million are opposite-sex couples. The remaining one million are same-sex couples, comprised of 462,215 male same-sex couples and

[6] Until 2013, people of the same sex who reported themselves as married in the American Community Survey were edited to be unmarried partners.

[7] In order to pin down partners in the household during the period around the entrance of the child, we could narrow our focus to married couples and use the date of marriage question in the ACS. However, that would not be a viable strategy because of the insufficient same-sex married couple sample size.

[8] Question 4 of the ACS asks about the age of the person and their date of birth. Since we are constraining our sample to couples, the couple accounts for two of the five persons about which detailed questions are asked.

[9] See Appendix A.1 for more details about the measurement of child timing for adopted children.

518,061 female same-sex couples. This means opposite-sex couples account for roughly 98.5% of couples, with male same-sex couples and female same-sex couples comprising 0.7% and 0.8% respectively.

Fisher et al. (2018) show that only 0.48% of joint tax filers are same-sex couples, lower than the ACS survey data rates. There are several reasons why we would expect survey data to yield a higher number of same-sex couples than joint tax filings. First, one member of the couple may not earn taxable income. Second, they may choose not to file jointly; they may have to file separately for their state taxes and choose to file federal taxes separately as well for logistical ease. They also may be following a practice of filing separate taxes that they established before 2013, when they could not file federal taxes jointly. Fisher et al. study tax filings from 2013 to 2015, a span that does not fully capture the additional time it takes couples to adjust their filing practices after the legalization of same-sex marriage, which did not happen at the federal level until 2015.

Fisher et al. (2018) also find that 7% of male same-sex joint tax filers and 28% of female same-sex joint tax filers have dependents. Interestingly, given the differences in the total percentages, the rate of same-sex couples with children in the tax data is roughly similar to the 10% of male and 22% female same-sex couples with children in the 2008-2018 ACS survey data (Badgett et al., 2021). This could be explained by joint tax filers overrepresenting couples with children, since there are several tax breaks that make filing jointly more beneficial for those with dependents. On the other hand, in couples with dependents, one of the partners is more likely to not have taxable income (Black et al., 2007).

In comparing our sample to the estimates from these other sources, it is important to note that our samples will differ from those estimates because we only include couples with children and couples who both match to LEHD employment data pre-parenthood. In addition, because we take a 10% sample of opposite-sex couples to create our main analysis sample, Table 1 is slightly misleading when it comes to estimating the proportion of same-sex couples. Before taking the random sample, opposite-sex couples account for 98.7% of couples, with 0.4% male same-sex

couples and 0.9% female same-sex couples. These estimates are similar to the published prevalence of same-sex couples in the ACS survey data and to the estimates based on our secondary (ACS) sample, which is 98.9% opposite-sex couples, 0.3% male same-sex couples, and 0.8% female same-sex couples. The fact that our estimates of the prevalence of same-sex couples with children among joint filers are higher than Fisher et al.'s estimates of same-sex couples among joint filers could suggest that, as discussed above, same-sex couples have been slowly adjusting their tax-filing conventions post-legalization of same-sex marriage.

3.4 Longitudinal Employer-Household Dynamics (LEHD) Data

We use data from the Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) to measure earnings. These data provide quarterly earnings records collected by state unemployment insurance (UI) programs, linked to establishment-level data from the Quarterly Census of Employment and Wages (QCEW). As noted earlier, LEHD data cover over 95% of private-sector workers and almost all state and local government employment (Abowd et al., 2009). We use LEHD data for all private-sector and public-sector employers from 2005-2021. The earnings measure we use throughout the paper is aggregate earnings from all employers in a given quarter and are real (2010) earnings. We winsorize earnings at the 99th percentile.

Our analysis sample is an unbalanced panel. We include births as early as 2005, the earliest year earnings are available to us, and as late as 2019, the last year of IRS filings available to us. We estimate 20 quarters (5 years) forward, so couples with children born in 2016-2019 will not be represented in all quarters. Analysis using a balanced panel, excluding those partially covered couples, gives qualitatively similar results.

3.5 Estimating the Earnings Path

We adopt the identification strategy used by Kleven et al. (2019), Andresen and Nix (2022), and other researchers estimating the child penalty. We rely on the exact timing of the event, namely, the quarter in which the first child enters the household, which creates a sharp discontinuity in

household production requirements. Our identifying assumption is there are no other factors that would cause a discontinuous change in earnings when the child arrives, except for the additional household production required by the arrival of the child itself.

$$y_{it} = \sum_{k=T_l}^{T_u} [\alpha_k \mathbb{1}(t - eventquarter_i = k)] + age_i + quarter_t + u_{it} \tag{1}$$

We estimate Equation 1 where the summation term represents a series of dummy variables indicating the distance in time from the quarter in which the first child arrives. We set the lower bound to be more than 2 years before the arrival of the child (quarter -9) and the upper bound to be more than 5 years after the arrival of the child (quarter 21). The "treatment" of becoming a parent is absorbing, so that those who have a child remain treated. This both matches our perception of reality and avoids the measurement error from using those who have already been treated as controls for those currently being treated, as discussed in Goodman-Bacon (2021). We omit quarter -6 so that our estimates represent total earnings relative to earnings a year and a half before the arrival of the first child. To control nonparametrically for life-cycle and time trends, we include age of parents at child arrival and fixed effects for each quarter. Our identifying variation then comes from variation in the ages of the parents and the exact quarters in which their children join the household. The disruption from having a child comes at different points in their age/earnings trajectory and at different calendar times.

For easier interpretation, we transform earnings levels into the percentage of the counterfactual earnings in the absence of a child. We use this approach instead of using log earnings to retain those in the sample who have zero quarterly earnings after the entrance of the child. Following Kleven et al. (2019), we define

$$P_t = \frac{\hat{\alpha}_t}{E[(\tilde{y_{it}}|t]} \tag{2}$$

where $E[\tilde{y_{it}}|t]$ represents the average predicted earnings from a regression without the event dummies, t refers to the event time (measured in quarters since arrival of first child), and $\hat{\alpha}_t$ is the estimated coefficient on the respective event time dummy taken from estimating Equation 1 above.

3.6 Defining Comparable Within-Couple Earner Types

We estimate Equations 1 and 2 separately for twelve different groups, as identified in Table 1. We first separate individuals into two groups based on their relationship status: opposite-sex couples and same-sex couples. This allows us to compare the earnings trends for these two types of couples, which is the focus of this paper. Our method for identifying same-sex individuals does not allow us to impute sexual orientation of uncoupled individuals, so we do not include uncoupled individuals in our analysis. Among each couple type, we further distinguish the partners by their pre-parenthood earnings, creating another three groups each.

We characterize individuals as the primary or secondary earner using their annual earnings from two years prior to the entrance of the child (that is, quarters -7 to -4). Primary earners are defined as those who make at least 10% more than their partner; secondary earners are those who make at least 10% less than their partner. Partners who make within 10% of each other in the period before the entrance of their first child are defined as equal earners. The size of these relative groups is shown in Table 1. In opposite-sex couples, male primary earners make up the largest group (904,000) and male and female equal earners make up the smallest group (89,500 each). In same-sex couples, female primary earners make up the largest group (131,000) and male equal earners make up the smallest group (6,600).

The higher earnings potential designation based on pre-parenthood earnings is preferred

over other approaches to differentiate between partners in same-sex couples such as using the Census-designated householder or the birth mother as the primary/secondary partner. The former is highly correlated with the earnings-based designation for opposite-sex couples but only minimally correlated for same-sex couples (Antecol and Steinberger, 2013), while the latter, though used by Moberg (2016) and Andresen and Nix (2022), is only applicable to female same-sex couples and does not address male same-sex couples or the many couples who adopt. A simpler version of our measure, without the equal earner category, has been used by Antecol and Steinberger (2013) to study earnings gaps among same-sex and opposite-sex couples; by Hansen et al. (2020) to investigate marriage premiums among gay and lesbian couples; by Martell and Nash (2020) to examine the effect of legalized marriage on hours worked; and by de la Vega (2022) to explore employment and earnings changes around childbirth among opposite-sex couples in Germany.

Table 2 is a transition matrix showing how many individuals move from being in a specialized couple to being in an equal-earner couple after the entrance of the child. We define pre-child specialization using the earnings in the one-year period starting one and a half years before the child enters; similarly, we define post-child specialization using the earnings in the one-year period starting one year after the child enters. For both same-sex and opposite-sex couples, the majority of individuals do not switch earner type. About 85% of same-sex couples and about 84% of opposite-sex couples remain specialized or remain equal earners.

If our measure of higher earnings potential was highly variant over time, this would raise concerns about its validity as a basis for differentiating between partners. However, the opposite is true: our measure is fairly constant over time, lending support to our use of this measure to proxy for pre-parenthood specialization. Though most do not change, there are some transitions between specialized and equal earner types brought on by the entrance of a child. Among

both opposite-sex and same-sex couples, slightly more individuals switch from specialized to equal than from equal to specialized after the entrance of the child (9.75% as compared to 5.87% for opposite-sex; 9.64% as compared to 5.54% for same-sex). We explore these shifts further through density plots in Section 4.

3.7 Propensity Score Weighting

Individuals in same-sex couples differ from those in opposite-sex couples in various ways as discussed in Section 2. Table 3 provides summary statistics for four groups: men in opposite-sex couples, women in opposite-sex couples, men in same-sex couples, and women in same-sex couples. In our main analysis sample based on tax and social security data, we have limited information about descriptive characteristics. However, we do see that women and men in same-sex couples are more likely to be Hispanic or Black than their counterparts in opposite-sex couples. Women in same-sex couples also earn more than their counterparts in opposite-sex couples, whereas men in same-sex couples earn about the same as their counterparts. On average, individuals in same-sex couples have fewer children (about 1.6 compared to 1.7 for opposite-sex couples) and are older when their first child enters the household (women are about 2 years older than their opposite-sex counterparts).

In the Appendix we present summary statistics for our second analysis dataset based on the ACS. As Table AT1 shows, this sample corroborates what we see in the administrative data; same-sex parents are older and more diverse, earn more, and have fewer children than opposite-sex parents. In addition, the ACS allows us to explore how education and marital status differ between the couple types. Men in same-sex couples are more likely to have a college degree than men in opposite-sex couples (62% versus 53%), whereas women are about equally likely to have

a college degree (61% versus 63%). However, women and men in opposite-sex couples are much more likely to be married, with about 90% of individuals in opposite-sex couples being married, compared to 78% of men and 65% of women in same-sex couples.[10]

Table 4 shows the pre-parenthood values of our main outcome variables, earnings and employment, disaggregated further into our twelve analysis subsamples defined by sex, couple type, and earner type. All earner types in same-sex couples earn, on average, at or above their counterparts in opposite-sex couples. Primary and equal earners of both sexes and couple types have employment rates at six quarters before the birth of over 90%, highlighting that this is a selected sample of parents that are generally highly attached to the labor force pre-birth. However, secondary earners are noticeably less connected to the labor force pre-birth. Secondary earner men in opposite-sex couples are the least likely to be employed pre-birth, at 69%. Appendix Table AT2 shows that the pre-parenthood earnings and employment in the ACS sample are similar to those measured in the administrative sample.

We have described several ways in which opposite-sex and same-sex couples differ based on characteristics that are observable in our data. For our analysis, we want to verify that any differences in earnings trends that we find are not due to same-sex couples simply being different from opposite-sex couples in pre-parenthood characteristics that are related to their income paths after their child's arrival. In other words, we want to ensure that same-sex couples and opposite-sex couples have similar background characteristics so that we are comparing couples whose income paths ought, other than for the introduction of a child, to be the same as one another. Therefore, we weight our main results by inverse propensity scores, which creates a sample

[10] Same-sex marriage was only available nationwide during part of our sample, with the Obergefell decision in 2015, which may explain much of this difference in marriage rates. The most recent cohorts, who had marital rights for a longer period before our sample period, may look more similar on this margin.

of opposite-sex couples that is more similar to our sample of same-sex couples.

We first predict the likelihood of being in a same-sex couple using the following observable characteristics of couples: the oldest age among the couple, the gap in ages within the couple, race, whether the couple is multi-racial, ethnicity, whether the couple is multi-ethnic, the gap in pre-child earnings, and the highest pre-parenthood income among the two. In our ACS sample, we additionally include the highest years of education, the gap in years of education, and marital status as predictors. Using this predicted probability, we create weights to give more predictive power to individuals who are more like individuals in the other type of couple. Opposite-sex couples who are more like same-sex couples are more highly weighted; same-sex couples who are more like opposite-sex couples are likewise more highly weighted. Specifically, the weight is calculated as the inverse of the predicted probability for same-sex couples and the inverse of one minus the predicted probability for opposite-sex couples.[11] Results that do not control for differences between same-sex and opposite-sex couples (that is, with no propensity weights) are presented in the appendix (see Figure AF2) and yield qualitatively similar results.

4 Results

We begin by presenting the percent change in earnings results for opposite-sex couples and same-sex couples. For each of the types of couples, we estimate the percent change in earnings after the entrance of the first child at the individual level, then extend the analysis to consider the impact at the household level. Next, we explore how the degree of specialization changed by comparing the gap in earnings before and after parenthood. We then present results on the in-

[11] For the secondary analysis sample, which is based on survey data, we combine these inverse propensity weights with ACS sample weights to form the final weights used in our regressions.

tensive and extensive margin responses. Finally, we discuss various ways in which our results are robust to observable differences between opposite-sex and same-sex couples.

4.1 Opposite-Sex Couples: Individual Earnings

Figure 1 Panel A shows the percent change in earnings after the entrance of the first child for opposite-sex couples. These earnings changes are relative to six quarters before the entrance of the child. The solid lines correspond to men and dotted lines correspond to women, while the color-symbol pairs (red circle, green square, and blue triangle) differentiate the earner types (primary, secondary, and equal). First focusing on only the difference between the male and female lines and not the earner types, we present support for the findings of the previous literature. Men in opposite-sex couples continue their upward pre-parenthood earnings trajectories, while women experience a significant decrease in earnings after the entrance of their first child followed by a partial recovery. One year after the child's entrance, women's earnings are about 20% lower than they were before the arrival of their first child and even after five years their earnings remain about 18% lower than before their first child. This child penalty has been well-documented in previous studies on the parental earnings gap.

We extend that literature by showing that the penalty exists across the split into earner types. All three types of opposite-sex female earners - primary, secondary, and equal -experience a decrease in earnings. In contrast, secondary earner opposite-sex men experience an increase in earnings beginning just before their child arrives, while primary and equal earner men see a consistent upward trend in their earnings that appears unchanged by the child entry.

The percent change in earnings is very similar in magnitude across earner types for women, with secondary earner women experiencing the most recovery, possibly due to their lower base-

line. The magnitude of the percent change in earnings varies more for men, with secondary earner men experiencing the largest increase relative to before the arrival of their child. By one year after the child joins the household, primary and equal earner men are earning 4% and 8% more than they were earning pre-child, respectively, whereas secondary earner men are earning about 35% more. That increase in earnings starts slightly before the child arrives; by the time the child enters the household, secondary earner men are already earning about 25% more than they were earning before the child.[12] By five years after, secondary earner men are earning about 52% more than they were before their first child, while primary earner men are earning about 10% more, and equal earner men are earning about 22% more.

We find no evidence of a fatherhood earnings premium, or a discontinuous increase in earnings around child arrival, among primary and equal earner men in opposite-sex couples, but some evidence among secondary earner men. This finding adds to the existing literature on the circumstances under which a fatherhood premium arises. While previous research has documented the existence of a fatherhood premium (Glauber, 2018; Pal and Waldfogel, 2016; Yu and Hara, 2021), much of that work uses either cross-sectional analysis or small samples, making it difficult to control for selection into fatherhood. Mari (2019) shows that the impact of parenthood on fathers' earnings varies by country with wage premiums found in the U.S. and Canada but not in Norway or Denmark. Our findings suggest that in addition to heterogeneity in fatherhood premiums by country, there is also heterogeneity by household earner dynamics. We find that the fatherhood premium exists for fathers whose female partners earned at least 10% more than they did before the arrival of their first child. de la Vega (2022) documents a similar pattern among

[12] We have explored whether this pre-child increase is a function of the period that we set as baseline (six quarters pre-child). Using four, five, and eight quarters instead all produce similar results.

opposite-sex couples in Germany. She shows that earnings and employment increase among men after childbirth, but that increase is concentrated in households where women were primary breadwinners. Similarly, Killewald (2013) uses panel data in the U.S. to explore heterogeneity in the fatherhood premium between different types of fathers and finds that the fatherhood premium exists only for married, residential, biological fathers whose wives do not remain full-time workers.[13]

Secondary earner fathers in our sample are significantly less attached to the labor force before the entrance of their first child, as Table 4 shows. We require individuals in our sample to have positive earnings at some point in the two years before the arrival of the child, but secondary earner men in opposite-sex couples are the least likely to be employed in our reference quarter (six quarters before the entrance of the child). 69% of secondary earner men in opposite-sex couples are employed in that quarter. In comparison, 94% of primary earner men in opposite-sex couples are employed. Even among secondary earners, men in opposite-sex couples are less often employed than women in opposite-sex couples (81%), men in same-sex couples (76%), and women in same-sex couples (75%). In addition, secondary earner men in opposite-sex couples are earning the least six quarters before the entrance of the child. They earn on average $5,100, less than half that of their primary earner counterparts ($12,500). That could explain why we see such an increase in earnings among secondary earner men in opposite-sex couples after child arrival. We explore this idea more in Section 6.

[13] Our current definition of primary earner does not differentiate between primary earners whose partners are not in the labor market and those that are. In future work as we explore labor force participation, we can look at this distinction and compare to the existing literature.

4.1.1 Opposite-Sex Couples: Household Earnings

At the household level, Figure 2 Panel A shows a similar drop in total labor market earnings at the time of child arrival for all three types of opposite-sex couples. However, male-primary couples experience a slightly smaller decline. In the first quarter after child arrival, their household earnings decrease by about 7% compared to before the child. Female-primary and equal-earner couples face a larger drop of about 16%. In all cases, earnings begin to recover two quarters after the first child arrives and fully recover by the end of our study period, five years later. By that time, equal-earner couples are earning approximately the same as before their child, male-primary couples are earning about 6% more, and female-primary couples are earning about 7% more. Based on the trends in individual earnings, it is likely that female-primary couples experience a quicker recovery of their earnings compared to equal-earner couples due to the large increase in earnings among secondary-earner men.

4.2 Same-Sex Couples: Individual Earnings

Figure 1 Panels B and C present the percent change in earnings after the entrance of the first child for female and male same-sex couples, respectively. We are able to identify precise estimates for same-sex couples thanks to our large administrative dataset, even for men in same-sex couples. Recall that our sample includes about 99,000 female same-sex couples and about 46,000 male same-sex couples. The precision of our estimates allows us to focus on the differences between the earnings trends for same-sex couples and those for opposite-sex couples, which we highlight here.

First, while men and women in opposite-sex couples have very different trends in earnings,

men and women in same-sex couples have very similar trends. This result was not necessarily expected, as previous literature suggests that male and female same-sex couples are different in ways that could be important to our analysis. For example, they enter parenthood at different rates; Badgett et al. (2021) reports that about 40% of individuals in opposite-sex couples have children, as compared to about 20% of females in same-sex couples and 10% of males in same-sex couples. They also have different patterns of division of household labor based on time-use research (Genadek et al., 2020). In our sample, men in same-sex couples earn more than women in same-sex couples (about $6,000 more on average) and are less white (43.4% non-white versus 33.9%). Nonetheless, the trends across sex of same-sex couples appear very similar.

Second, in contrast to opposite-sex couples, we observe differences in same-sex couples when we look across earner types. For all earner types, there is a drop in earnings after the entrance of the child for both male and female individuals in same-sex couples of around 4 to 16%, which is small relative to the 45% decline we see for opposite-sex women, but also does not follow the pattern of increasing earnings seen in opposite-sex men. However, after the initial drop, earnings trends diverge for primary, secondary, and equal earners. Same-sex secondary earners experience an increase in long-run earnings relative to pre-parenthood, while primary and equal earners have no change in long-run earnings relative to pre-parenthood. By five years after the entrance of the first child, male and female secondary earners earn 14 to 19% more than they did before the child. In contrast, male and female equal earners earn about 1% more, and male and female primary earners earn about 1% less.

These results suggest a kind of equalization may be occurring in same-sex couples, since primary earners do not, on average, increase their earnings while secondary earners do. We explore this further in the transition matrix in Table 2. Around 9.6% of those who were specialized

before the child would be categorized as equal earners after the child, which provides further evidence that same-sex couples who are specialized before the child on average move toward equalization after the child. In addition, same-sex couples overall are more likely to be equal earners after parenthood than before (11% versus 6.9%). On the other hand, about 5.5% of those couples who were equal-earners before switch to being specialized after. Figure 1 uses only pre-child earnings to define the earner categories for the analysis, so that the categories are not changing across treatment. This means that some of the individuals in the equal earner group would actually be categorized as primary or secondary earners based on post-child earnings. Because both of the individuals in a given equal-earner couple contribute to our estimate of the trend in earnings for that group, when one individual increases earnings and the other decreases earnings, those trends could net out to the zero change in earnings that we see in Figure 1. The combined evidence from the event studies and transition matrices highlights that the overall effect of parenthood on equalization of earnings versus specialization in earnings among same-sex couples is unclear. We explore this further by analyzing how the distribution of the income gap between individuals changes after parenthood in Section 4.3.

4.2.1 Same-Sex Couples: Household Earnings

Figure 2 presents the results for household earnings. These figures show similar changes in household earnings across same-sex versus opposite-sex couples, despite the differences at the individual level. Same-sex couples, like opposite-sex couples, experience a drop in household earnings of 8 to 14% one quarter after their first child enters the household. By the second quarter, the drop lessens to about 5% less than earnings before the first child. By the end of the sample period (five years after the first child), household earnings have fully recovered. In opposite-sex

couples, the household earnings trends we see are a result of increases in male earnings which just about offset the decreases in female earnings, regardless of earner type. The household earnings trends for same-sex couples follow a similar pattern to those of opposite-sex couples, but the earnings trends within the couples that create these aggregate household trends are very different. Instead of a gap in earnings between individuals that aggregates to zero change at the household level, the relatively constant household earnings in same-sex couples appear to be the result of an equalization of earnings between different earner types within the couple.

4.3 Distribution of the Gap in Earnings

Figures 3 and 4 present kernel densities characterizing the earnings gap between the two individuals in the household for opposite-sex couples and for same-sex couples. The earnings gap is defined as the difference between the earnings of the primary earner and those of the secondary earner. Our classification for primary, secondary, and equal earners is the same used throughout this paper: we categorize individuals based on their annual earnings from two years before the child's arrival. When individuals are equal earners, we set one to be the "primary" for this analysis so that the position of the two individuals stays constant (that is, individual "secondary" is always subtracted from individual "primary"). We measure the earnings gap at two points: before the child's entrance (quarter -4) and after the child's entrance (quarter +4). The dotted blue line shows the distribution of the gap in earnings before parenthood and the red solid line shows the distribution after the child enters the household.

Starting with opposite-sex couples, for male-primary-earner couples (Panel A of Figure 3), in almost all cases the earnings gap is positive, meaning the primary earner earns more than the secondary earner. The distribution is right-skewed; in some couples, the male primary earner

earns significantly more than his spouse. But there is still a mass near zero, where the primary and secondary earners are earning close to the same amount (even with equal earners excluded from the sample; Panel C presents the earnings gaps for equal earners). After the child is born, the distribution shifts to the right, meaning primary earners are earning more relative to their partners than they were before the child was born. The shift of mass away from zero indicates more specialization.

For opposite-sex couples in which the female spouse was the primary earner before the child was born (Panel B), the pre-birth distribution also shows right skewness, but there are fewer cases of women earning significantly more than their male partners. In the post-birth distribution, we see less mass near zero, indicating more specialization. But for female primary earners, that specialization is driven more by their male partners becoming more connected to the labor force. The increased mass is almost entirely on the negative side of the distribution, meaning the male spouses now earn more than the female primary earners do, despite those men having earned less than their partners in the pre-child period.

For equal-earning spouses in opposite-sex couples (Panel C), there is a large mass at zero, as expected since equal earners are defined as earning within 10% of each other in the pre-birth period. After the child is born, specialization increases in these couples, and the mass at zero is much smaller than before, with mass moving to either tail.

Figure 4 presents the results for same-sex couples. For male same-sex primary and secondary earners (Panel A), the pre-birth distribution has a long right tail similar to that of male-primary-earner opposite-sex couples, showing that some male primary earners earn significantly more than their partners. There is also a large mass near zero, indicating that there are many male same-sex couples in which the partners earn close to the same amount. After birth, the mass near

zero decreases and we see movement into the left tail as some secondary earners begin to earn more than their primary partners. However, we also see movement into the right tail. The movement in both tails suggests that the shift toward specialization is not driven uniformly by only primary earners or only secondary earners taking on more work outside the house, but instead that both cases are occurring. For equal-earning male couples (Panel B), like in equal-earning opposite-sex couples, the mass at zero decreases after birth, indicating more specialization.

Female same-sex primary and secondary earner couples (Panel C) look similar to male same-sex primary and secondary earner couples. The distribution is right skewed with a large mass near zero. In the post-birth distribution, the mass near zero decreases and shifts toward both the left and right tails, showing that specialization has increased for couples in both directions. This contrasts with the female-primary-earner opposite-sex couple figure (Panel B of Figure 3), in which the parenthood specialization is driven by increased earnings of the secondary earner (male). As with the other couple types, we also see greater specialization among equal-earning female partners (Panel D) after birth.

Overall, all couple types show increased specialization after a child is born, regardless of the pre-birth earnings distribution. However, the most uneven shifts to specialization happen in opposite-sex couples. Female-primary-earner opposite-sex couples are the only group where all of the movement is into the left tail, with no movement into the right tail. Male-primary-earner opposite-sex couples exhibit only movement into the right tail, with primary earners further expanding their earnings advantage. Whether specialized or not pre-birth, we see an increase in specialization within same-sex couples, but there is movement into both tails; either spouse, whether primary or secondary earner pre-birth, might become the primary earner post-birth.

4.4 Intensive and Extensive Margins

To understand the contribution of changes in employment to the overall changes in earnings, we analyze the extensive and intensive margins of earnings. Specifically, we examine the percentage change in earnings due to changes in employment versus the percentage change in earnings from within employment. To achieve this, we replace quarterly earnings with the individual's pre-child earnings whenever the quarterly earnings are zero. This technique transforms our variable measuring changes in earnings from a negative to a zero value in any period where the individual has no earnings, relative to the baseline earnings level. This method is akin to that used by Groen et al. (2020) in their analysis. This substitution of earnings for zero values only works for levels of earnings, not percentage changes, so the results discussed in this section are in levels.

Figure 5 displays two lines: the change in total earnings for the entire sample and the change in earnings for the intensive margin. The gap between the total earnings and the intensive margin earnings is the extensive margin earnings, or the loss in earnings from those who experienced a change in their employment status from baseline.

For primary earner men in opposite-sex couples (Panel A), we observe an increasing trend in earnings after the entrance of their child. However, this increasing trend would have been even higher if some men had not left their jobs during that period. Earnings one year after the child for primary earner men are $1,200 higher if just including the intensive margin, but only $300 higher for all earnings. At 5 years the gap is larger, with growth to $3,700 at the intensive margin and only $1,500 for all earnings. The earnings of equal earner men also exhibit an increasing trend that would have been even higher if not for those who experienced periods of zero earnings post-

child. In contrast, the total earnings line is above the intensive margin line for secondary earner men reflecting that secondary earner men are more likely to be employed after their child's arrival than at baseline. The increase in their earnings is partially due to an increase in earnings on the intensive margin and partially due to an increase in employment.

For women in opposite-sex couples, we observe a dip in both the total and intensive margin earnings after child arrival. However, the intensive margin earnings losses are smaller, $1,400 versus $3,300 losses measured for all earnings for primary earner women. They also recover to the previous level after just a couple of quarters such that primary earner women have had earnings growth at the intensive margin of $500 after one year, whereas the extensive margin earnings do not recover and pull down the total earnings level to an average loss of $1,800 after one year. The pattern holds for women in opposite-sex couples, regardless of whether they are the primary, secondary, or equal earner in the couple, suggesting that a major contributor to the depressed earnings of women in opposite-sex couples after child arrival is decreased employment.

In comparison, men and women in same-sex couples (Panel B) have small drops in earnings after becoming parents along both the intensive and extensive margins. But after the first or second quarter, we see increasing trends in earnings at the intensive margin for all six groups. For primary and equal earners, there is a larger gap between total and intensive margin earnings with larger losses due to a change in employment, in comparison to secondary earners for whom there appears to be a smaller change in employment status. For primary earners, the gap between intensive and extensive margin earnings is around $2,000 for both men and women five years after their child, while for secondary earners it is around $1,300.

4.5 Robustness Checks

We make several decisions in defining our analysis sample and method that could conceivably influence our results. Here we explore in more detail several of those decisions. We first present results without propensity score weighting, then results that focus only on likely non-adoptive parents. Next we turn to our secondary analysis sample based on ACS data, and use that sample additionally to explore the impact of using a different control group. Across all these analyses, we find results that are very similar to the main results we discuss above.

4.5.1 Without Propensity Score Weighting

We have so far shown that when a child enters the household, the earnings of women and men in opposite-sex couples respond differently than the earnings of women and men in same-sex couples. However, we have also shown in Table 3 (summary statistics; see Section 3 for a discussion) that women and men in opposite-sex couples are different from women and men in same-sex couples on many dimensions. Thus another interpretation of our results could be that the differences in earnings trends we identify are due to different rates of marriage, different rates of educational attainment, or different likelihoods of having a second child in the household. If that were the case, then after controlling for those differences, we should see no difference in the earnings trends of individuals in opposite-sex and same-sex couples.

In our main results, we use the propensity weights discussed in Section 3 to create a sample of opposite-sex couples that are similar to same-sex couples on observable characteristics. This allows us to argue that the differences we observe between the two groups are due to the gender mix of the couples rather than to other factors. We perform two additional exercises to evaluate

the influence of observable characteristics, both of which suggest that they have little influence on the differences we see.

First, in the Appendix we provide results without using propensity weights. Figure AF2 shows that our results remain almost identical. For individuals in both female and male same-sex couples, the unweighted earnings trends are slightly less positive than the weighted earnings trends, but these differences exist only at the third decimal place. The similarity between our results with and without weights suggests that the difference in earnings trends that we see are not driven by differences in characteristics of individuals in same-sex versus opposite-sex couples.

Second, we add controls one by one to our specifications to see which variable is driving the differences between opposite-sex and same-sex couples. We consider similar controls to those that create the propensity score: age, difference in age, race, difference in race, ethnicity, difference in ethnicity, pre-child earnings, household earnings, and age of the child upon entrance to the household. We find that no variable makes a notable difference in the results. This is consistent with our comparison of earnings trends with and without propensity score weighting. Just as the propensity score did not significantly affect the overall patterns, the results with the controls added separately also show no differences in the patterns we identify.

4.5.2 Adoptive parents

Adoption is another dimension on which opposite-sex and same-sex couples differ. In Appendix Figure AF3, we explore whether the difference in earnings trends can be explained by the differential rates of adoption among opposite-sex and same-sex couples. We first intended to limit our sample to only those parents who first claimed their child at least one year after the child was born. However, delays in claiming children on tax forms are driven not only by adoption, but also

by other factors such as changes in family structure, changes in tax filing, and lacking information about the criteria to claim dependent children. Instead, we limit our sample to those children who were likely *not* adopted; we eliminate families in which the child was first claimed later than their birth year. Kleven et al. (2019) found that adopting parents experience a similar long-term gender gap in earnings as biological parents, and our results confirm this finding. Our results are consistent with this; Appendix Figure AF3 shows that likely non-adopting parents have very similar trends to the full sample of parents, suggesting that adoption cannot explain the differences in earnings we observe between opposite-sex and same-sex couples.

While there are some small differences between the likely-non-adoptive parent and full sample results, qualitatively the patterns are very similar. Individuals in male and female same-sex couples experience a drop in earnings after the child enters, but their earnings quickly recover fully, with secondary earners seeing the largest increase in earnings relative to before the child. There is suggestive evidence that non-adoptive parent secondary earners in female and male same-sex couples see slightly less of an increase in earnings after the child enters the household than their full-sample counterparts, but the confidence intervals do not reject that the trends are the same.

Like our main results, males in opposite-sex couples do not see a drop in earnings around the quarter in which the child enters, but instead see a steady positive trend in earnings. Secondary earner males in opposite-sex couples see the largest increase in earnings. Like for same-sex couples, there is some visual evidence that adoptive parent secondary earners experience a slightly smaller increase in earnings after the child enters the household than their full-sample counterparts, but the differences are not statistically distinguishable.

4.5.3 Secondary Analysis Sample: ACS

As we discuss in detail in Section 3, there are several downsides to relying on a combination of Social Security and tax records to identify our couples. As an alternative, we instead create a sample based on couples identified in the American Community Surveys (ACS). This approach allows us to include both married and unmarried couples, and gives us more demographic information on which to base our propensity score weights. Comparing the results from these two different samples helps us understand whether the differences we see in the earnings trends are driven by differences in observable characteristics of same-sex and opposite-sex couples, especially marital status.

We present results from the ACS sample in the Appendix. Figure AF4 presents the results most comparable to our main analysis, using propensity weights and expressing changes in percent terms relative to the pre-child period. The results are strikingly similar to those based on the administrative data. The most dramatic difference is that the confidence intervals around the male and female same-sex couple estimates are much larger, owing to the much smaller sample available in the ACS. However, the point estimates from the ACS sample are very much aligned with those from the administrative sample. For all couple types, earnings changes are slightly less extreme – smaller declines, smaller growth – but not statistically distinguishable from their comparable administrative data estimate. We see more evidence of anticipation among secondary earners in our administrative results, with pre-childbirth earnings increases beginning around the 5th quarter before the child enters, compared to the 3rd quarter in the ACS. Overall, though, comparing our results across the two datasets suggests that the trends in earnings are consistent across marital status and not driven by demographic characteristics of the couples.

4.5.4 Future parents as control group

In all our specifications we treat the entrance of a child as an absorbing state, so already treated parents do not serve as controls for those who are not yet treated. In Appendix Figure AF5, we provide a robustness check that uses parents who have a child after our observation window as a control. We do this by shortening our observation period: we only include earnings up to 2014, although we continue to use births through 2019 from the American Community Survey (ACS). Parents whose children enter their households between 2014 and 2019 serve as controls for individuals who become parents before 2014. We argue that this is a better control group than individuals who never become parents. As in our main specification, we include couple type, age, and calendar quarter fixed effects so that the earnings effects are measured relative to individuals of similar age and couple type over time. We find that the patterns of earnings after the child's arrival remain similar when we use this control group.

5 Theoretical Framework for Household Choice after Entry of First Child

We develop a theoretical framework for understanding household choices made around the arrival of the first child to understand the patterns found in our empirical exercises more deeply. Our focus is on decisions made after the entry of the child, so we do not model differences in preferences for having children. Since we do not have information on the identity of the birth parent, we do not model differences based on this important, but omitted, characteristic.[14] We also do not explicitly model any additional constraints that same-sex couples may face in becoming

[14]Like opposite-sex couples who face fertility issues, same-sex couples also face biological issues that can affect which, if either, of the members of the couple will be biologically related to the child and, for female same-sex couples, if either member of the couple carries the child through pregnancy. For couples who adopt, the timing and age of the child at adoption are also potential choice variables.

parents, which may differ from those faced by opposite-sex couples.[15]

We define two types of people, male (M) and female (F), who divide their time between labor production and household production. They are given initial productivity draws in both of these markets, with productivity in the labor market (wages) denoted w_i and productivity in the household denoted η_i. Each person receives a random initial productivity draw for the two markets from a normal distribution with means $\overline{w_M} > \overline{w_F}$ and $\overline{\eta_M} = \overline{\eta_F}$. Productivity increases annually in both markets based on the number of hours dedicated to that market in the previous period.

$$w_{it} = w_{it-1} + \gamma * w_{it-1} L_{it-1}$$

$$\eta_{it} = \eta_{it-1} + \phi * \eta_{it-1} H_{it-1}$$

The individuals are subject to a time constraint, such that the number of hours devoted to labor (L) and household (H) production cannot exceed their total number of hours, which we set to 60 per week.

A household, before the entrance of the first child, consists of two people (persons 1 and 2). They face a household production constraint such that the two individuals in the couple must jointly produce a fixed amount of household good $G_h = G*$. Household good production is subject to diminishing marginal returns. Total household goods are defined by

$$G_h = \eta_1 t * H_1^{\alpha} + \eta_2 t * H_2^{\beta} \text{ where } 0 < \alpha < 1, 0 < \beta < 1$$

[15]Some of these additional constraints reflect existing laws that treat parental rights differently for same-sex couples. For example, while all 50 states allow same-sex couples to adopt children, 17 states have no explicit protections against discrimination based on sexual orientation or gender identity, and 12 states allow state-licensed child-welfare agencies to refuse to place children in families if doing so conflicts with their religious beliefs (Project). In addition, since same-sex couples are less likely to live near their places of birth (Black et al., 2007), they may be less able to rely on extended family resources for childcare, limiting their choice set.

Subject to these constraints, the household maximizes consumption (total household labor production) in the periods before the child arrives.

Two changes occur after the child arrives in the household:

1. The amount of household production required ($G*$) permanently increases;

2. Productivity growth (ϕ) for hours spent in the household sector increases in only the year of child arrival. In other words, learning by doing for these new household production tasks is very high.

In addition to maximizing household income, we explore three other approaches couples could take to allocate hours between household and labor sectors: specialization based on earnings, equalization of household labor hours, and specialization based on gender norms.

We can compare all three of these allocation strategies to the "maximizing income" strategy to understand the potential income loss from following one of these alternative strategies. The patterns from our simulated theoretical couples can also be compared to the patterns found in the observed couples in our data to understand the potential approaches real couples are taking to make these decisions.

We simulate our model using a synthetic set of 10,000 individuals divided evenly into men and women, with half in opposite-sex couples and the other half in same-sex couples, so that we have 2,500 opposite-sex couples and 1,250 each of male same-sex and female same-sex couples. Initial productivity draws are from normal distributions with $\overline{w_M} = 20, \overline{w_F} = 15, \sigma_{wM} = \sigma_{wF} = 5$ and $\overline{\eta_M} = \overline{\eta_F} = 10, \sigma_{\eta M} = \sigma_{\eta F} = 1$.

The initial amount of household good required to be produced by the household ($G*$) is 30 units; that amount increases in period 4, when the child enters the household, to 60 units. The

values of α and β, which drive the decreasing returns to household production hours for each person, are set to 0.5. Productivity in both the household and labor markets increases at 5% a year for full-time work in either sector (defined as 8 hours for the labor market and 4 hours for the household market) and increases more slowly for less than full-time and more quickly for more than full time (ie $\gamma = 0.05 * L_{it-1}/8$ and $\phi = 0.05 * H_{it-1}/4$). The temporary increase in productivity growth for hours devoted to the household sector in the birth year is 10% for full-time work in that sector, and more for additional hours ($\phi = .10 * H_{it-1}/4$).

Figure 6 shows the individual earnings trajectories, depicted as changes in dollars earned, for the couples based on the maximizing household income decision rule and the three alternative decision rules, with couples grouped in the same way as in the empirical results. The four decision rules depicted are (1) maximize household income, (2) specialization based on comparative advantage earnings, (3) equal division of household labor hours, and (4) specialization based on gender norms. Scenario 4 only applies to opposite-sex couples.

We find that only when the division of childcare responsibilities is driven by gender norms (Scenario 4) would we observe the significant decrease in earnings of all female partners in opposite-sex couples (including those with higher earning potential than their male partners) that we see in the data. This result aligns with Weeden et al. (2016) who argue that the wage gap is perpetuated by social norms that lead to gender-based differences in hours worked, and consequently earnings. Furthermore, Kleven (2022) shows that the average child penalty varies across space in the U.S. in line with local social norms around gender roles.

Our empirical results are inconsistent with the model predictions for Scenarios 2 and 3. Scenario 2 (comparative advantage in earnings) predicts that primary earners would not see a drop in earnings while secondary earners of both sexes would, but empirically we see that pri-

mary earner women do experience a drop in their earnings and secondary earner men do not. Scenario 3 (equal allocation) predicts that all sexes and earner types would see a small drop in earnings, whereas empirically we see that opposite-sex men of all earner types do not see a drop in their earnings and opposite-sex female earners see a large drop.

One inconsistency between the model and the observed outcomes is the pattern of earnings for secondary earner men in opposite-sex couples. In Scenario 4 of the model (gender norms), the earnings growth for secondary earner men is lower than that for primary earner men, because the earnings growth is entirely based on initial earnings. However, in the data we see that secondary earner men earn more to compensate for the loss of household income when their higher-earning partners experience a child penalty.

The results for same-sex couples suggest an equal allocation (Scenario 3) of post-child household production and reject specialization based on potential earnings (Scenario 2). However, the model does not fully capture some of the patterns seen in the data, particularly the secondary earner growth and the lack of growth for the primary earner. This may suggest there are more complicated household bargaining dynamics at play that are not captured by our simple model. Our empirical results that disaggregate the effects into intensive and extensive margins show some of those dynamics. In same-sex couples, the secondary earners are not adjusting their labor at the extensive margin; the intensive margin and total earnings lines run nearly parallel. However, the primary earners in same-sex couples are making extensive margin adjustments. We also see extensive margin adjustments among equal earners in same-sex couples, which explain why we see some shifting from the equal earner group to the specialized group in Table 2. This could be evidence that same-sex couples are specializing according to work preferences if those preferences are not strongly correlated with the earnings categories.

6 Discussion and Future Work

We leverage a uniquely large and detailed data source to characterize changes in earnings for parents in several different couple types. Notably, this represents some of the first evidence on the earnings of same-sex parents in the United States. We show that same-sex couples, as compared to opposite-sex couples, experience different earnings trajectories after a child enters their household. In opposite-sex couples, we find robust evidence of a child penalty applying almost exclusively to the female partner, even if she was the primary earner before parenthood. In same-sex couples, however, neither partner experiences as strong or as persistent a decline in earnings. This difference in earnings trajectories suggests the different couple types follow different decision rules on how to allocate the additional time burden of caring for a young child. Specialization in childcare and market work, with the choice of who specializes driven strongly by gender norms, seems to be the typical response of opposite-sex couples, while we see more complicated dynamics in the response of same-sex couples.

Same-sex couples face the same challenges as opposite-sex couples in the U.S., including lack of affordable childcare, universal parental leave, and mandated sick leave, but same-sex couples may also face additional challenges which we have not captured. Moreover, we have made the simplifying assumption that work-life preferences across opposite-sex and same-sex couples are the same. Keeping in mind these potentially confounding factors, it is noteworthy that same-sex couples allocate their time differently than opposite-sex couples. These differences in behavior over the sex composition of couples are consistent with Kleven's (2022) findings that gender norms are one of the most important drivers of the child penalty for mothers, both in the U.S. and likely worldwide.

We have endeavored to bring as much data as possible to our motivating question, but there remain many avenues for possible future research. As noted above, we made several simplifying assumptions about both constraints and preferences. As more data become available for the relatively small same-sex couple populations, it may be possible to extract more information about constraints through geographic variation in laws and policies differentially impacting same-sex and opposite-couples with children. Qualitative surveys and research in other disciplines may provide additional insight into the preferences underlying these time allocation decisions.

7 Tables

Table 1: Sample Sizes

Opposite-Sex Couples					
1,637,000					
Male Primary	Male Secondary	Male Equal	Female Primary	Female Secondary	Female Equal
904,000	310,000	89,500	644,000	559,000	89,500
Male Same-Sex Couples			Female Same-Sex Couples		
65,800			138,750		
Male Primary	Male Secondary	Male Equal	Female Primary	Female Secondary	Female Equal
62,500	33,500	6,600	131,000	74,500	15,500

Note: This table presents the number of observations in the analysis sample, broken apart by couple type, sex, and earner type. Same-sex couples are determined by jointly-filing couples from the IRS and reported sex of the individual and their partner from the Numident. Equal earners make within 10% of their partner's earnings; when the gap in earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. The individual types in rows two and four that make up a couple do not always have the same number of observations because we require each observed person to have nonzero earnings in the pre-child period. Thus, for example, the number of Male Primary individuals in Opposite-Sex Couples is larger than the number of Female Secondary individuals in Opposite-Sex Couples because some of the partners of Male Primary individuals have zero earnings pre-child and do not enter our analysis.

Table 2: Specialization

	Opposite-Sex Couples			Same-Sex Couples		
	Specialized Pre	Equals Pre	Total	Specialized Pre	Equals Pre	Total
Specialized Post	82.99	5.87	88.86	83.44	5.54	88.98
Equals Post	9.75	1.39	11.14	9.64	1.38	11.02
Total	92.74	7.26	100.00	93.08	6.92	100.00
N	2,141,000			277,000		

Note: This table presents an evaluation of how much the earner type designation changes over time. Our designation categorizes individuals based on their earnings in the "pre" period defined as 6 and 9 quarters before the birth of the child. When the gap in earnings is 10% or less, both members of the couple are designated as equal earners; when the gap in earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. The "post" version instead uses earnings from 5 to 9 quarters after the child is born to categorize earner types. The resulting tabulation shows how many individuals change categories after they have their first child. Specialized refers to individuals who are either primary or secondary earners.

Table 3: Summary Statistics

	Opposite-Sex Couples		Same-Sex Couples	
	Men	Women	Men	Women
Age at Birth	30.74	28.59	30.47	29.76
	6.471	5.85	6.724	6.853
Age at Entrance	31.31	28.84	31.13	30.69
	6.5	5.927	7.003	7.055
Age Child at Entrance	.5065	.2239	.5832	.84
	1.604	1.051	1.718	2.061
Number of Children	1.708	1.711	1.679	1.525
	.7928	.7931	.8028	.7372
Prebirth Earnings	40890	30040	40700	34830
	40800	31170	46800	37700
Primary Earner	.732	.501	.6272	.6214
	.5298	.5725	.5544	.5618
Secondary Earner	.3482	.5798	.4453	.4582
	.5503	.5724	.5643	.572
Equal Earner	.08019	.08086	.0725	.07963
	.2716	.2726	.2593	.2707
Non-White	.3192	.3169	.4338	.3392
	.4662	.4653	.4956	.4734
Observations	969,000	961,000	77,500	170,000

Note: This table provides descriptive statistics (mean and standard deviation) about the analysis sample. We divide the sample into couple types (opposite-sex and same-sex) and further into men and women. All variables come from the Numident except for earnings (from the LEHD) and primary, secondary, and equal earner indicators (created using LEHD, IRS, and Numident information).

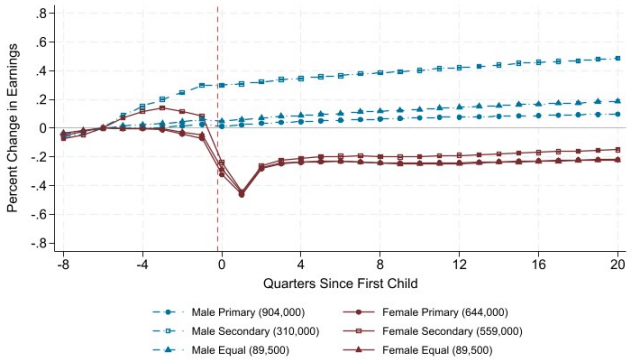
Table 4: Prebirth Earnings and Employment

	Opposite-Sex Couples					
	Male Primary	Male Secondary	Male Equal	Female Primary	Female Secondary	Female Equal
Average Earnings	12,500	5,100	11,000	9,300	5,700	11,000
Percent Employed	94.3%	69.3%	96.6%	92.7%	81.1%	98.0%
	Male Same-Sex Couples			Female Same-Sex Couples		
	Male Primary	Male Secondary	Male Equal	Female Primary	Female Secondary	Female Equal
Average Earnings	13,000	6,000	12,500	11,000	5,400	11,500
Percent Employed	93.3%	75.7%	97.2%	92.9%	75.1%	97.0%

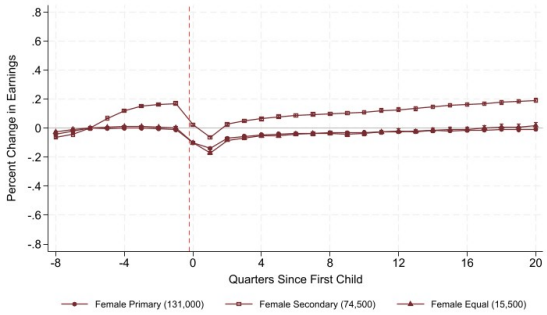
Note: This table provides descriptive statistics on the pre-parenthood values of our main outcome variables. The values are earnings and employment measured 6 quarters before the entrance of the child into the household. The sample is divided into the twelve analysis groups defined by couple types (opposite-sex and same-sex), sex (male and female), and pre-birth specialization (primary, secondary, and equal earners).

8 Figures

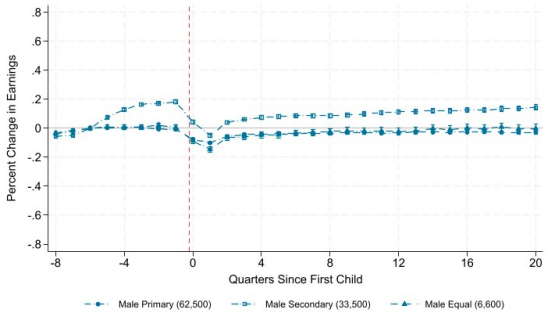
Figure 1: Individual Earnings after Entry of First Child



(a) Individuals in Opposite-Sex Couples



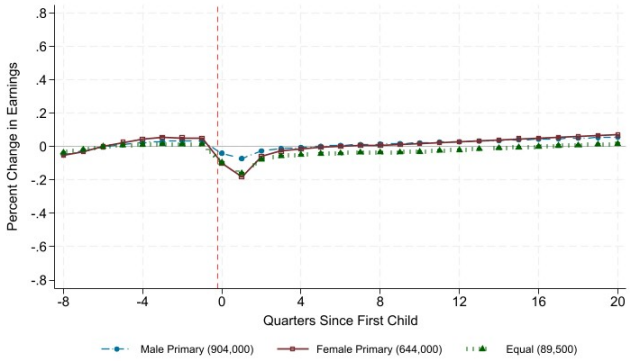
(b) Individuals in Female Same-Sex Couples



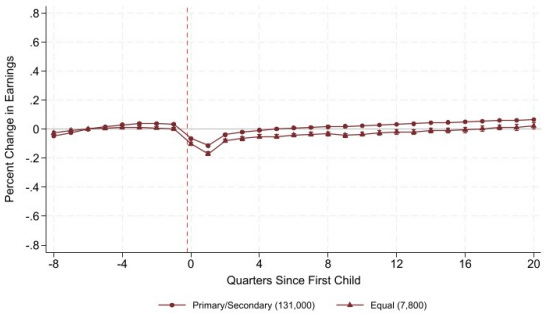
(c) Individuals in Male Same-Sex Couples

Note: This figure presents the main event studies of earnings around child entry at the individual level. For easier interpretation, earnings changes are presented as percent changes following Kleven et al. (2019). The results are weighted with propensity weights to address differences in underlying characteristics of opposite-sex and same-sex couples. When the gap in pre-birth earnings is 10% or less, both members of the couple are designated as equal earners; when the gap in pre-birth earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. Individuals in the analysis sample are required to have nonzero earnings in at least one quarter in the two years before the child enters the household. Source: Numident, IRS, and LEHD data.

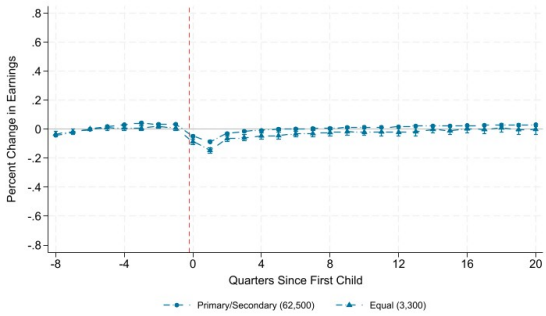
Figure 2: Household Earnings after Entry of First Child



(a) Individuals in Opposite-Sex Couples



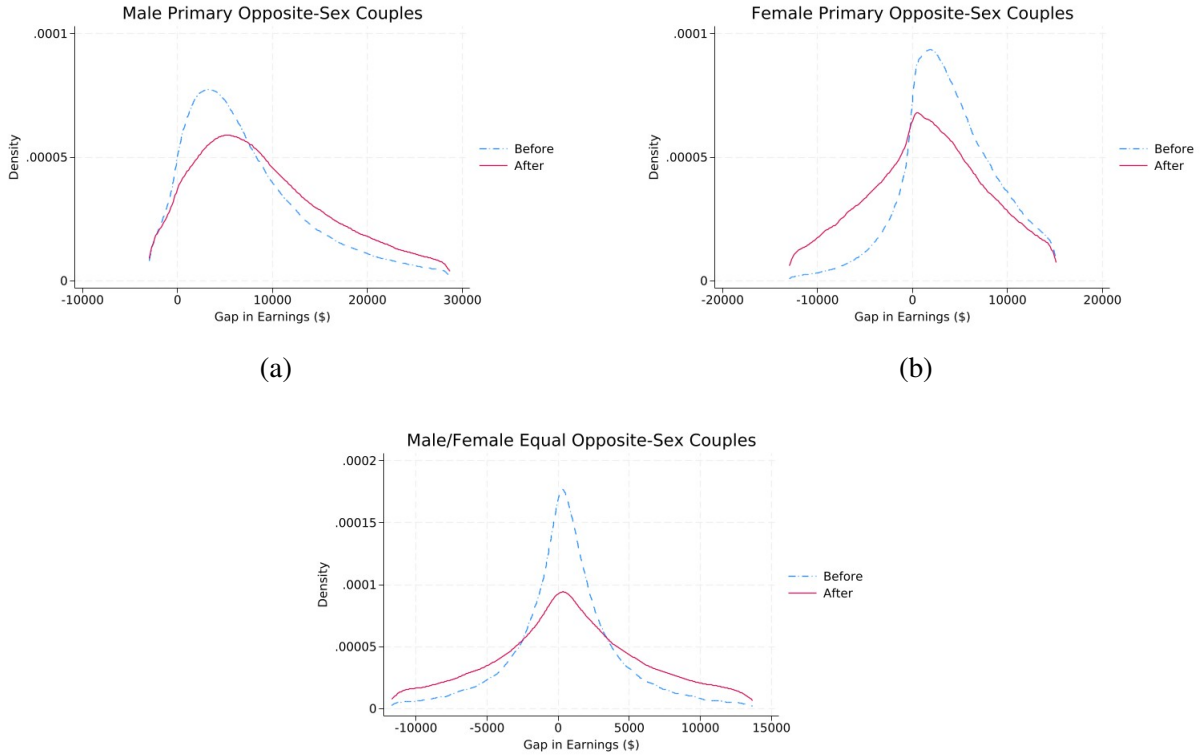
(b) Individuals in Female Same-Sex Couples



(c) Individuals in Male Same-Sex Couples

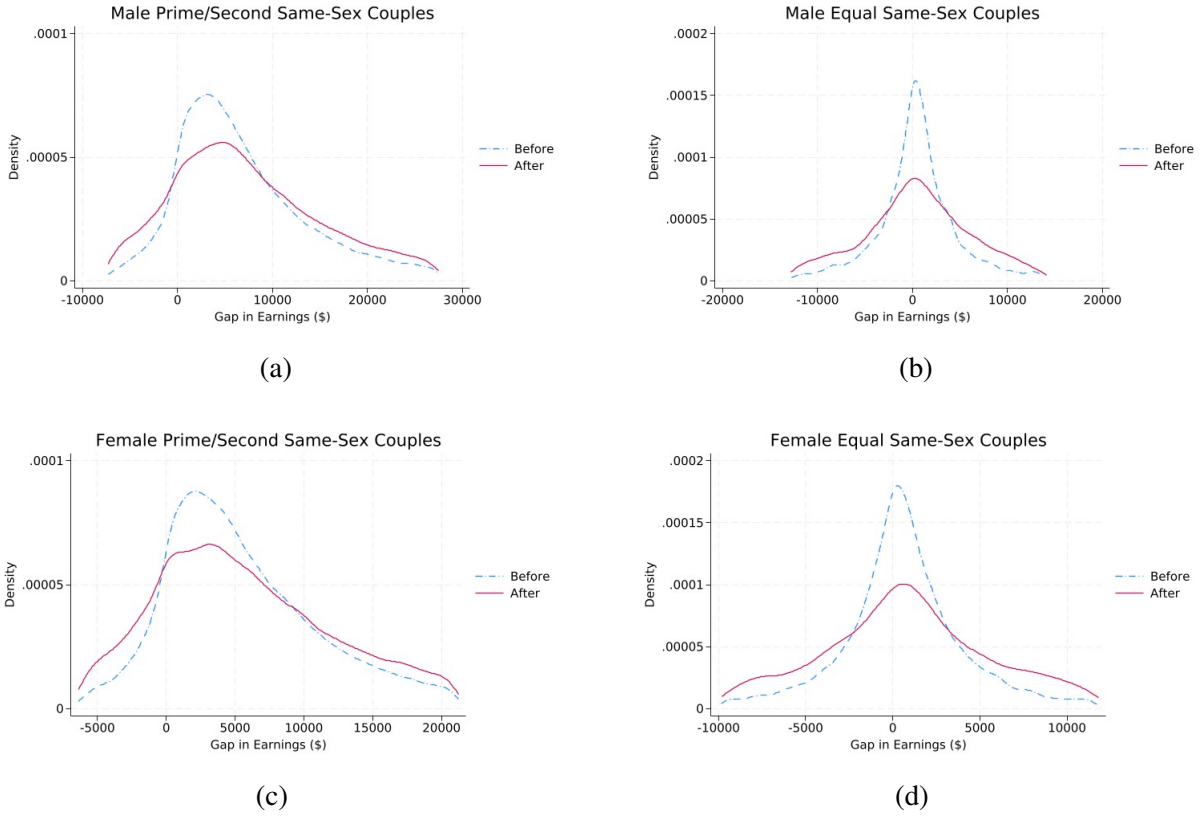
Note: This figure presents the household-level event studies of earnings around child entry. For easier interpretation, earnings changes are presented as percent changes following Kleven et al. (2019). The results are weighted with propensity weights to address differences in underlying characteristics of opposite-sex and same-sex couples. When the gap in pre-birth earnings is 10% or less, both members of the couple are designated as equal earners; when the gap in pre-birth earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. This analysis sample is slightly less restrictive than the individual-level analysis sample because we include individuals with no pre-birth earnings as long as their household has nonzero pre-birth earnings in at least one quarter. Source: Numident, IRS, and LEHD data.

Figure 3: Distribution of Gap in Earnings for Opposite-Sex Couples Before and After First Child



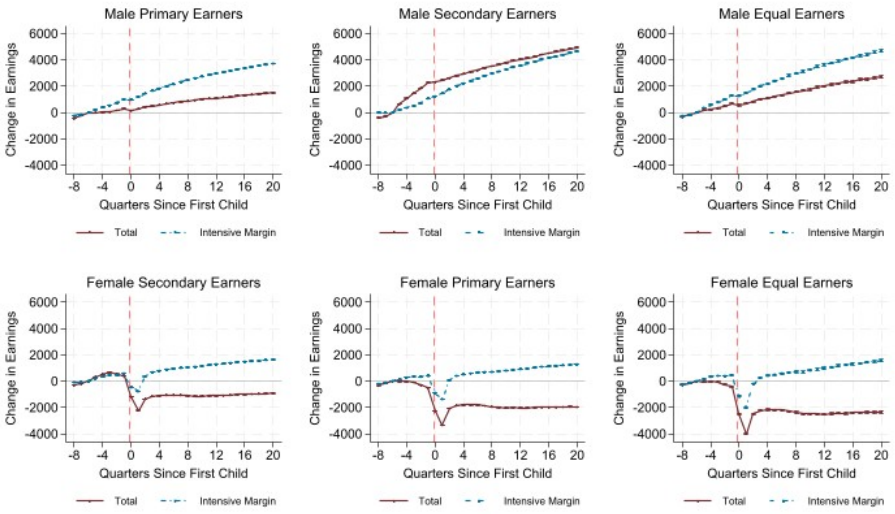
Note: This figure presents kernel densities of the distribution of the gap in earnings for different opposite-sex couple types. Before refers to the year before the child enters the household, and after refers to the year after the child enters. The gap is calculated as $earnings_{prebirthhigherearner} - earnings_{prebirthlowerearner}$. When the gap in pre-birth earnings is 10% or less, both members of the couple are designated as equal earners; when the gap in pre-birth earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. Individuals in the analysis sample are required to have nonzero earnings in at least one quarter in the two years before the child enters the household. Source: Numident, IRS, and LEHD data.

Figure 4: Distribution of Gap in Earnings for Same-Sex Couples Before and After First Child

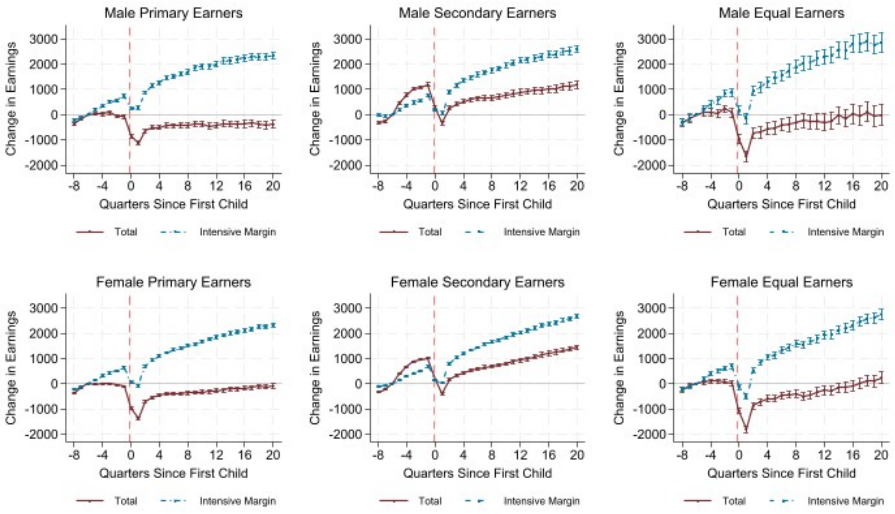


Note: This figure presents kernel densities of the distribution of the gap in earnings for different same-sex couple types. Before refers to the year before the child enters the household, and after refers to the year after the child enters. The gap is calculated as $earnings_{prebirthhigherearner} - earnings_{prebirthlowerearner}$. When the gap in pre-birth earnings is 10% or less, both members of the couple are designated as equal earners; when the gap in pre-birth earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. Individuals in the analysis sample are required to have nonzero earnings in at least one quarter in the two years before the child enters the household. Source: Numident, IRS, and LEHD data.

Figure 5: Individual Earnings after Entry of First Child, Intensive Margin



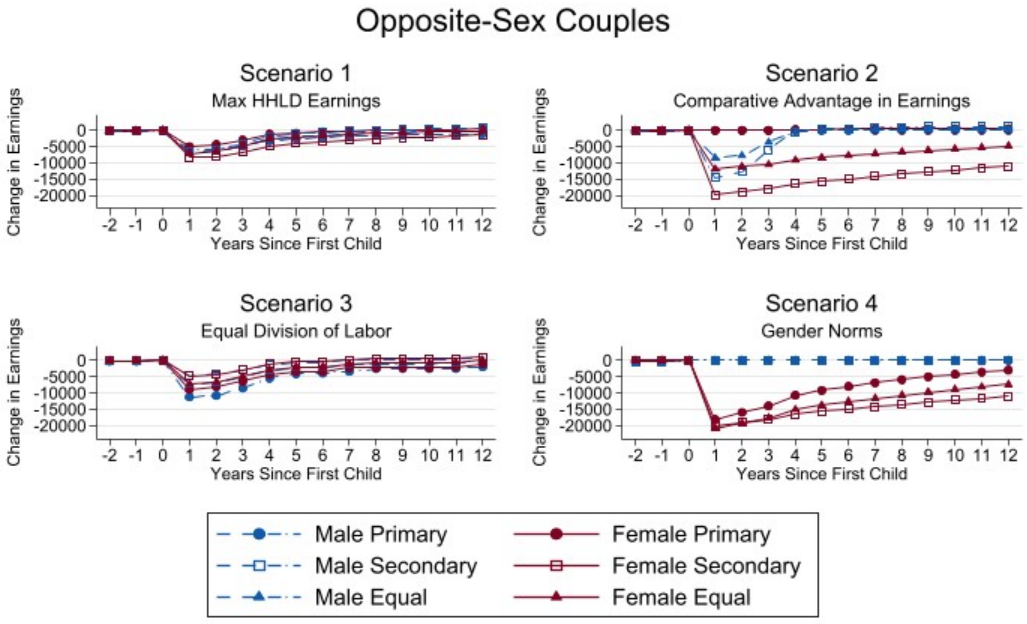
(a) Individuals in Opposite-Sex Couples



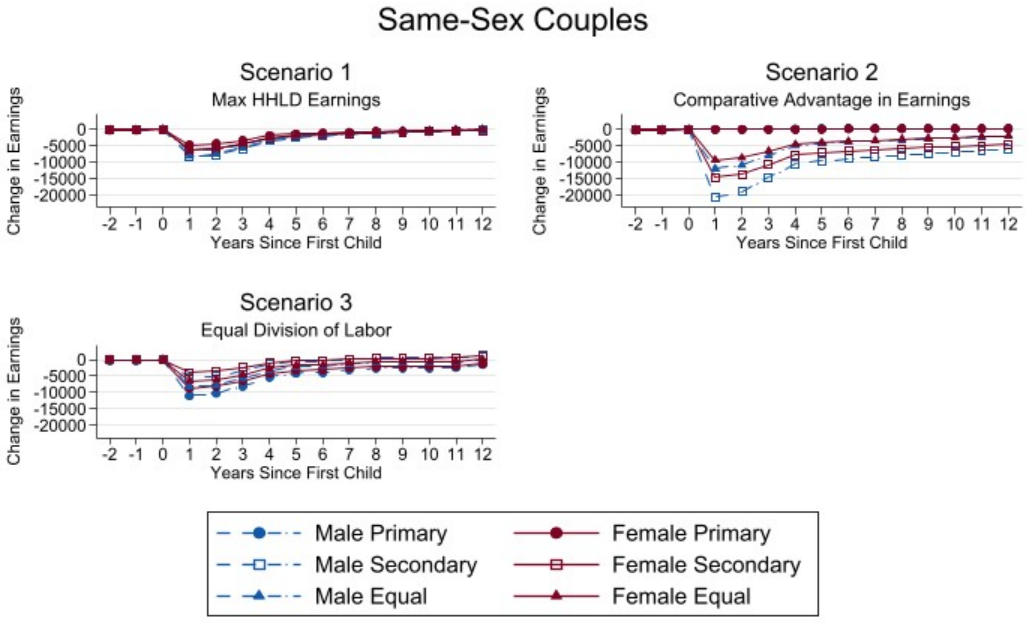
(b) Individuals in Same-Sex Couples

Note: This figure presents a decomposition of the event studies of earnings around child entry at the individual level into the intensive and extensive margins. The results are weighted with propensity weights to address differences in underlying characteristics of opposite-sex and same-sex couples. When the gap in pre-birth earnings is 10% or less, both members of the couple are designated as equal earners; when the gap in pre-birth earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. Individuals in the analysis sample are required to have nonzero earnings in at least one quarter in the two years before the child enters the household. Source: Numident, IRS, and LEHD data.

Figure 6: Theoretical Model Predictions



(a) Individuals in Opposite-Sex Couples



(b) Individuals in Same-Sex Couples

Note: This figure presents the predicted earnings path around child entry based on simulations of the theoretical model. The scenarios explore couples dividing their time either (1) by maximizing household income, (2) based on comparative advantage in earnings, (3) based on equalization of hours for home production between members of the couple, or (4) based on gender norms. See Section 5 for a full description of the model.

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A Appendix

A.1 Measurement Details

A.1.1 Identifying Same-Sex Couples

The ACS collects detailed relationship information relative to Person 1 for up to 4 other people in the household through question 2. Up through 2018 in our sample, question 2 has 15 checkbox responses. Starting in 2019, question 2 has 16 responses. Figure AF1 below shows the question 2 for both 2013 and 2019 and provides the instructions from 2013. To summarize, the following changes occur between 2013-2018 and 2019: (1) Husband-wife is disaggregated into opposite-sex and same-sex; (2) Unmarried partner is disaggregated into opposite-sex and same-sex and appears earlier in responses; (3) Roomer or boarder is dropped; and (4) ordering of phrase "Housemate or roommate" is switched.

A number of papers address the measurement of same-sex couples in available U.S. data. Starting with survey data, Badgett et al. (2021) compare results from the ACS to results from the National Health Interview Survey (NHIS) and Behavioral Risk Factor Surveillance System (BRFSS) and note consistencies across the sources. The NHIS and BRFSS are notable in that they have a question asking about sexual orientation, whereas in the ACS same-sex status can only be inferred by the sexes of the individuals in a couple until 2019 when it is explicitly collected. The NHIS (2013-2018) identifies 1.4% of women as same-sex and 1.8% of men; in comparison the ACS (2008-2018) identifies 1.2% of women as same-sex and 1.1% of men. When limiting the ACS to years 2013 and beyond, the share of all couples who identify as same-sex is between 1.2% and 1.6%; in comparison the BRFSS (2014-2018) identifies 1.7% of all couples as same-sex.

Turning to administrative data, Fisher et al. (2018) combine tax records from 2013 to 2015 with the SSA Numident for information on sex and find that 0.48% of joint tax filers are same-sex, or about 59% of the number of same-sex partners identified in the ACS. There are many possible reasons for this mismatch, including misreporting on the survey, inaccuracies in the sex re-

ported in the Social Security information, and differences in the likelihood to file jointly. Fisher et al. argue that the most compelling reason for the mismatch is the difference in how marriage is defined for tax versus survey purposes. In particular, in states where same-sex marriage had just been legalized, couples might not have been legally married on their tax returns but might identify as being in a marriage-like relationship, or might have been able to legally marry by the time of the survey.[16]

A.2 Identifying Child Timing

Note that we do not have birth records, unlike Moberg (2016) and Andresen and Nix (2022), who use data from several Scandinavian countries. Thus we cannot determine which of two female same-sex partners gave birth. In a similar fashion, we do not know if one of the partners in a same-sex male couple is genetically related to the child. In the ACS data, all relationships are defined relative to the reported head of household. In our main results we combine three categories of children: biological children, adopted children, and stepchildren, and do not attempt to separately identify the birth and non-birth parent.

Several studies have provided information on the differences between same-sex couples and opposite-sex couples in terms of the timing, amount, and likelihood of having children. Using the 2019 ACS, Walker and Taylor (2021) find that same-sex couples have far fewer children under 18 present in the household than opposite-sex couples (about 59% of opposite-sex couples have no children present, while about 86% of same-sex couples have no children present). Badgett et al. (2021) come to similar conclusion using both the 2008-2018 ACS and 2013-2018 NHIS. From the ACS they calculate that about 44% of women in opposite-sex couples have children in the household (38% of men), whereas only 22% of women in same-sex couples have children (10% of men). Using the NHIS they can identify same-sex individuals even when they are

[16]The counts tabulated from the three surveys and the administrative tax data discussed above pertain only to adults in relationships. For an overall tabulation of individuals identifying as same-sex in the United States, Gates (2010) finds that about 3.5% of adults identify as lesbian, gay, or bisexual. He uses the National Epidemiological Survey on Alcohol and Related Conditions (2004-2005), National Survey of Family Growth (2006-2008), General Social Survey (2008), California Health Interview Survey (2009), and National Survey of Sexual Health and Behavior (2009) for his calculations.

not in a couple, so their calculations are for all adults rather than only those in couples. Therefore the categories they define are slightly different than those used in our analysis; they find that 33% of straight women have children (31% of men), while 20% of lesbian women have children (5% of gay men).

Since adoption is an important component of family formation for same-sex couples, we needed to correct for potential timing differences between the child's date of birth and the date the child joined the ACS household we observe them in. We first identified the population of adopted children in the 2013-2019 ACS using relationship type 3 (adopted child) to the head of household. We only kept those who could be matched to Census data, or had a Protected Identity Key (PIK) for both the parent (either household head or spouse) and the child. We then cleaned up duplicates, which mostly seemed to consist of the same household being surveyed in the ACS multiple times in the 2013-2019 span. In cases of duplicate parent-PIK and child-PIK pairs, we kept the earliest ACS record. But we otherwise did not deduplicate on child or adult PIK, so duplicate child PIKs remain when they match to more than one parent PIK and duplicate parent PIKs remain when they match to more than one adopted child. We used records from up to 6 children in each ACS household, starting with the eldest.

We then merged the parent PIKs from the ACS adopted children file to the IRS 1040 data for years 2000-2018. We did this as a many-to-many match, since there were duplicates on both sides. On the ACS side, duplicates occurred because there were duplicates parent PIKs when parents had more than one adopted child; on the IRS side, duplicates occurred when the parent PIK had multiple 1040 records.

For those records where a parent PIK matched, we then did a match to the child PIK, creating an indicator if the child PIK matched any of the (up to) 6 PIKs on the 1040 record. We then deduplicated the records within year. If there were 1040 records for a PIK that both included the child and did not include the child, we chose the ones with the child PIK. We also chose records where the spouse PIK appeared rather than ones where the spouse PIK did not. After those two targeted deduplications, we randomly dropped the remaining duplicates of parent-PIK, child-PIK

pairs within year.

We then merged all of the IRS years 2000-2018 together, assigned a first year observed to each child-PIK, parent-PIK pair to get a year from the IRS data. To find an estimated age at adoption we combined the child-birth year from the Census Numident, if available (or the ACS date of birth if not) with this first year observed in the IRS. If the parent PIKs never appear in the IRS 1040 data, then both the parent and the child will not appear in this final file, and the ACS birth date is then used as a default, instead of the estimated adoption date.

A.3 Additional Tables and Figures

Table AT1: Summary Statistics: ACS Sample

	Opposite-Sex Couples		Same-Sex Couples	
	Men	Women	Men	Women
Age at Entrance	30.31	28.26	32.33	30.83
	(6.346)	(5.805)	(7.501)	(7.682)
Age of Child at Entrance	0.044	0.031	0.368	0.226
	(0.527)	(0.446)	(1.696)	(1.242)
Adopted Child	0.024	0.024	0.289	0.210
	(0.153)	(0.153)	(0.454)	(0.407)
Number of Children	1.903	1.902	1.750	1.649
	(0.895)	(0.894)	(0.843)	(0.827)
Prebirth Earnings	94,230	68,710	111,700	84,440
	(88,010)	(68,390)	(116,300)	(92,650)
Primary Earner	0.721	0.486	0.623	0.610
	(0.540)	(0.575)	(0.570)	(0.564)
Secondary Earner	0.364	0.598	0.468	0.469
	(0.558)	(0.574)	(0.583)	(0.572)
Equal Earners	0.085	0.085	0.090	0.079
	(0.278)	(0.278)	(0.287)	(0.270)
Married	0.905	0.905	0.782	0.647
	(0.293)	(0.293)	(0.413)	(0.478)
Black	0.056	0.046	0.060	0.080
	(0.229)	(0.210)	(0.238)	(0.271)
Hispanic	0.095	0.094	0.126	0.108
	(0.293)	(0.293)	(0.332)	(0.311)
Less than HS	0.053	0.031	0.056	0.043
	(0.225)	(0.174)	(0.231)	(0.202)
HS	0.414	0.338	0.327	0.353
	(0.493)	(0.473)	(0.469)	(0.478)
College	0.533	0.631	0.616	0.605
	(0.499)	(0.483)	(0.486)	(0.489)
Observations	1,345,000	1,350,000	3,800	11,000

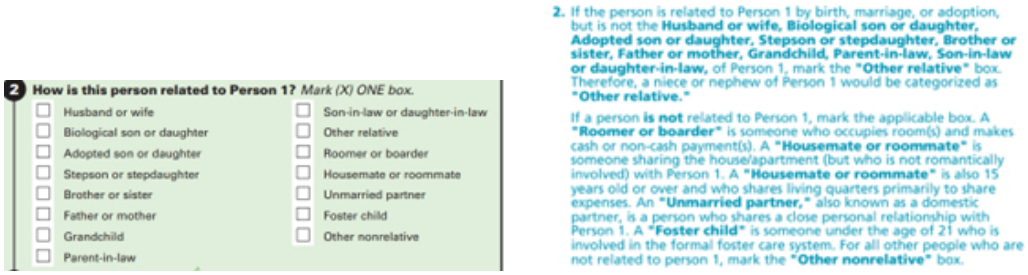
Note: This table provides descriptive statistics (mean and standard deviation) about the analysis sample. We divide the sample into couple types (opposite-sex and same-sex) and further into men and women. All variables come from the ACS except for earnings (from the LEHD) and primary, secondary, and equal earner indicators (created using LEHD, IRS, and Numident information).

Table AT2: Prebirth Earnings and Employment: ACS Sample

	Opposite-Sex Couples					
	Male Primary	Male Secondary	Male Equal	Female Primary	Female Secondary	Female Equal
Average Earnings	12,250	5,206	10,460	9,253	5,559	10,340
Percent Employed	94.1%	68.2%	96.2%	92.3%	80.5%	97.6%
	Male Same-Sex Couples			Female Same-Sex Couples		
	Male Primary	Male Secondary	Male Equal	Female Primary	Female Secondary	Female Equal
Average Earnings	15,270	7,321	13,410	11,390	5,610	11,420
Percent Employed	94.5%	79.0%	96.4%	92.9%	73.6%	95.8%

Note: This table provides descriptive statistics on the pre-parenthood values of our main outcome variables. The values are earnings and employment measured 6 quarters before the entrance of the child into the household. The sample is divided into the twelve analysis groups defined by couple types (opposite-sex and same-sex), sex (male and female), and pre-birth specialization (primary, secondary, and equal earners).

Figure AF1: Relationship Question in the ACS 2013 and 2019



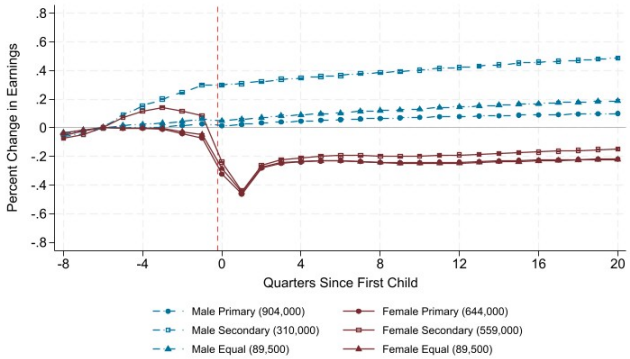
(a) 2013



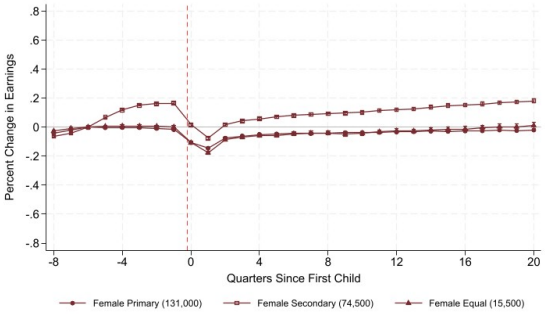
(b) 2019

Note: This figure shows how the relationship question was asked in the ACS in years 2013 (Panel A) and 2019 (Panel B). Source: ACS website `https://www.census.gov/programs-surveys/acs/about/forms-and-instructions.html`

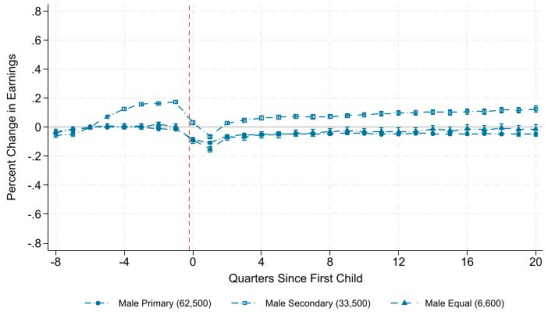
Figure AF2: Individual Earnings after Entry of First Child, Without Propensity Weights



(a) Individuals in Opposite-Sex Couples



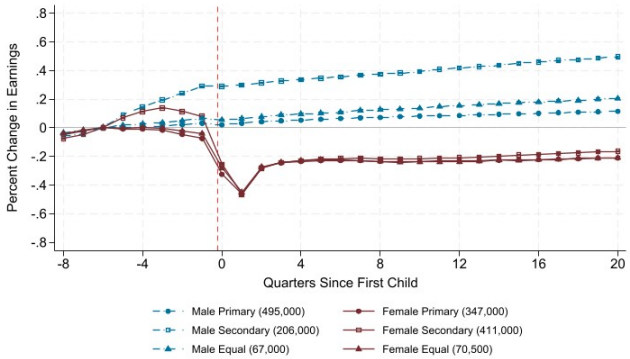
(b) Individuals in Female Same-Sex Couples



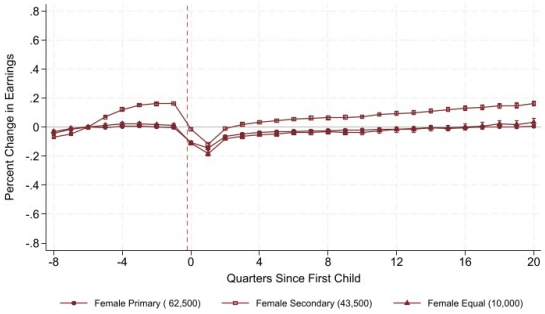
(c) Individuals in Male Same-Sex Couples

Note: This figure presents the main event studies of earnings around child entry, but the results do not use propensity weights. For easier interpretation, earnings changes are presented as percent changes following Kleven et al. (2019). When the gap in pre-birth earnings is 10% or less, both members of the couple are designated as equal earners; when the gap in pre-birth earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. Individuals in the analysis sample are required to have nonzero earnings in at least one quarter in the two years before the child enters the household. Source: Numident, IRS, and LEHD data.

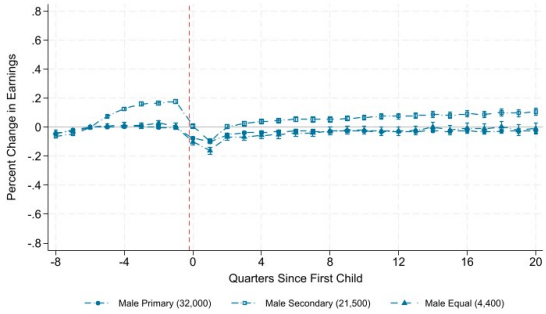
Figure AF3: Individual Earnings after Entry of First Child, Among Likely Non-Adoptive Parents



(a) Individuals in Opposite-Sex Couples



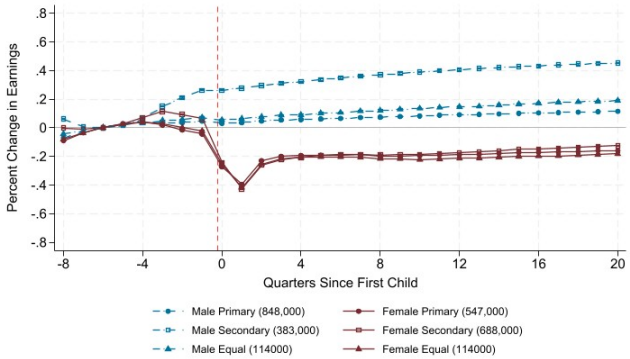
(b) Individuals in Female Same-Sex Couples



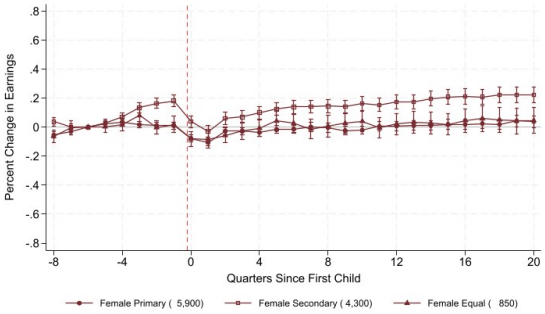
(c) Individuals in Male Same-Sex Couples

Note: This figure presents the main event studies of earnings around child entry at the individual level, but limiting the sample only to parents whose first child was likely not adopted (born same year as first claimed on parent's taxes). For easier interpretation, earnings changes are presented as percent changes following Kleven et al. (2019). The results are weighted with propensity weights to address differences in underlying characteristics of opposite-sex and same-sex couples. When the gap in pre-birth earnings is 10% or less, both members of the couple are designated as equal earners; when the gap in pre-birth earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. Individuals in the analysis sample are required to have nonzero earnings in at least one quarter in the two years before the child enters the household. Source: Numident, IRS, and LEHD data.

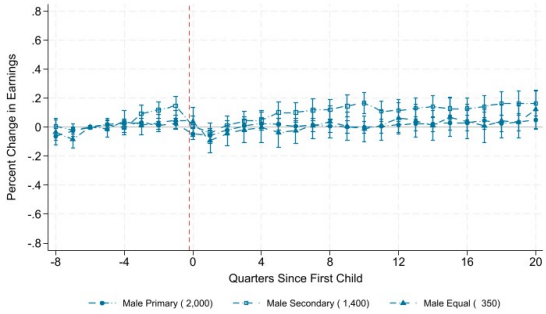
Figure AF4: Individual Earnings after Entry of First Child, ACS Sample



(a) Individuals in Opposite-Sex Couples



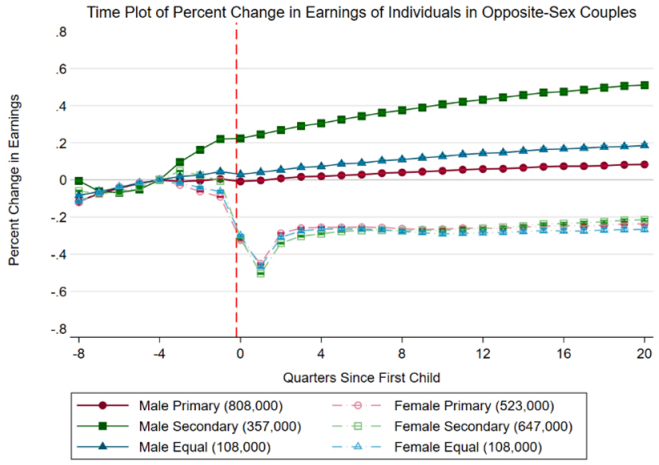
(b) Individuals in Female Same-Sex Couples



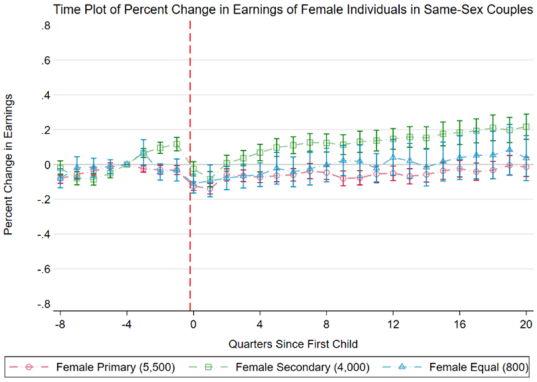
(c) Individuals in Male Same-Sex Couples

Note: This figure presents the main event studies of earnings around child entry, but the results are based on the ACS sample (see Section 3). For easier interpretation, earnings changes are presented as percent changes following Kleven et al. (2019). When the gap in pre-birth earnings is 10% or less, both members of the couple are designated as equal earners; when the gap in pre-birth earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. Individuals in the analysis sample are required to have nonzero earnings in at least one quarter in the two years before the child enters the household. Source: ACS, Numident, and IRS data.

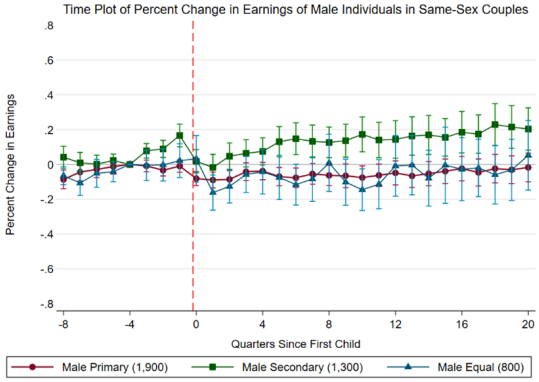
Figure AF5: Individual Earnings after Entry of First Child, Parents-to-be As Controls



(a) Individuals in Opposite-Sex Couples



(b) Individuals in Female Same-Sex Couples



(c) Individuals in Male Same-Sex Couples

Note: This figure presents the main event studies of earnings around child entry, but the results are based on the ACS sample and only use earnings up to 2014. Parents whose children enter the household between 2014-2019 then serve as the control group, comparing with parents whose children enter the household before 2014. For easier interpretation, earnings changes are presented as percent changes following Kleven et al. (2019). When the gap in pre-birth earnings is 10% or less, both members of the couple are designated as equal earners; when the gap in pre-birth earnings is larger than 10%, earners are designated as primary if they have higher earnings than their partner and secondary if they have lower. Individuals in the analysis sample are required to have nonzero earnings in at least one quarter in the two years before the child enters the household. Source: ACS, Numident, and IRS data.